NC STATE UNIVERSITY

ECE 592-091 - Resonant Power Converters

Lab 2 - Prototyping and Open-Loop Testing of a Class D Series-Resonant Inverter (SRI)

Sanjana Ashok Bhalekar
Raghav Chandrasekaran
Sydney Cheng
Abhishek Ajit Deshpande

Professor - Dr. Zeljko Pantic
Teaching Assistance - Zhansen Akhmetov

Table of Contents

Table of Contents **2**
Introduction **3**
Task 1: Measuring the frequency characteristics of the inductor **4**
Aim 4
Test Setup 4
Procedure Conducted 5
Observations and Results 6
Task 2: Resonant network assembling and tuning **15**
Aim 15
Test Setup and Procedure 15
Observations and Results 16
Task 3: Measuring the inverter's DC-AC voltage gain and efficiency **28**
Aim 28
Test Setup 28
Procedure Conducted 28
Observations and Results 29
Task 4: Testing ZVS vs. Hard switching **36**
Aim 36
Procedure Conducted 36
Observations and Results 36
Conclusion **39**

Introduction

Lab 2 focuses on developing a deeper understanding of frequency characteristics in passive components and their role in resonant circuits. Students will begin by measuring the frequency response of inductors and capacitors and tuning a resonant network. The lab then progresses to assembling a series-resonant inverter, where students will experimentally determine its DC-AC voltage gain function (MVI) and efficiency (ηI). Additionally, the lab explores key differences in switching waveforms, allowing students to observe and compare zero-voltage switching (ZVS) and hard-switching transitions in real-world applications.

Task 1: Measuring the frequency characteristics of the inductor

Aim

In Task 1, the LCR meter was employed to analyze the frequency characteristics of the inductor. Specifically, the instrument was utilized to determine the equivalent high-frequency model of the inductor, as outlined in Topic 1 of the course and illustrated in Figure 1.1. This process involved measuring the inductor's response to varying frequencies, allowing for a detailed assessment of its behavior in high-frequency applications.

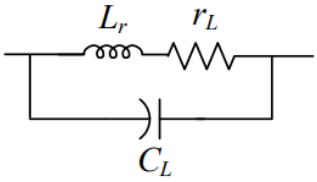


Fig. 1.1: High-frequency model of the resonant inductor

Test Setup

- Performed short- and open-circuit tests using the Adjustment (ADJ) tab of the LCR meter to determine measurement error corrections. Before testing, the meter's cable length was set to 1 m.
- Connected the inductor to the LCR meter and linked the meter to a computer via USB cable. Rejected the request to reset the instrument to prevent loss of compensation values. In case of an accidental reset, refer to Figure 1.2 for the required settings.

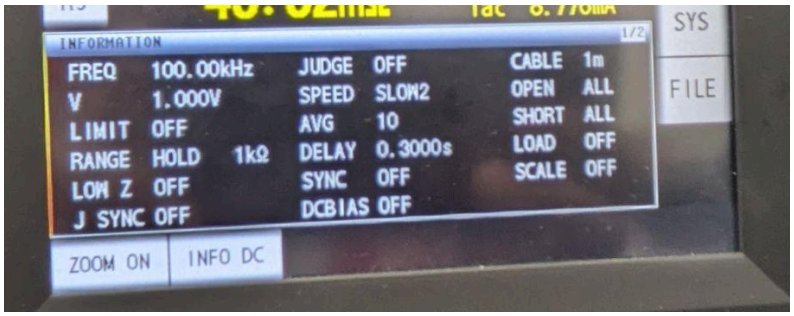


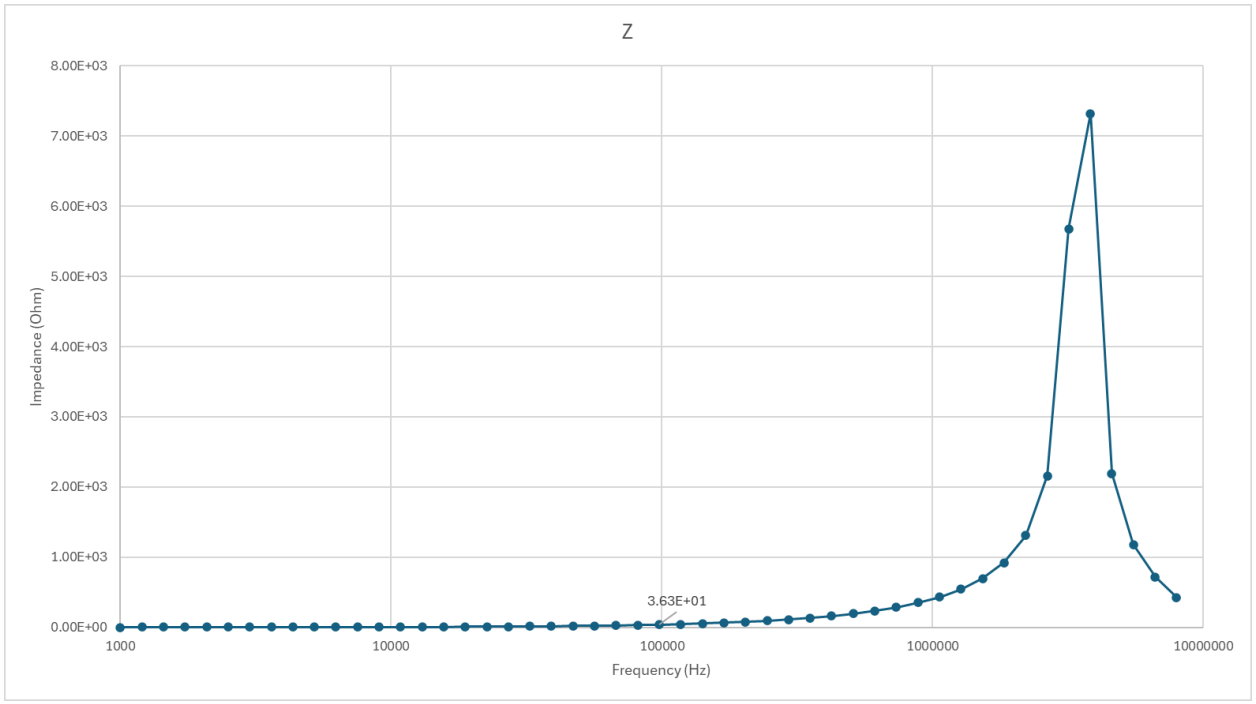
Fig. 1.2: Experiment setting based on LCR meter required settings

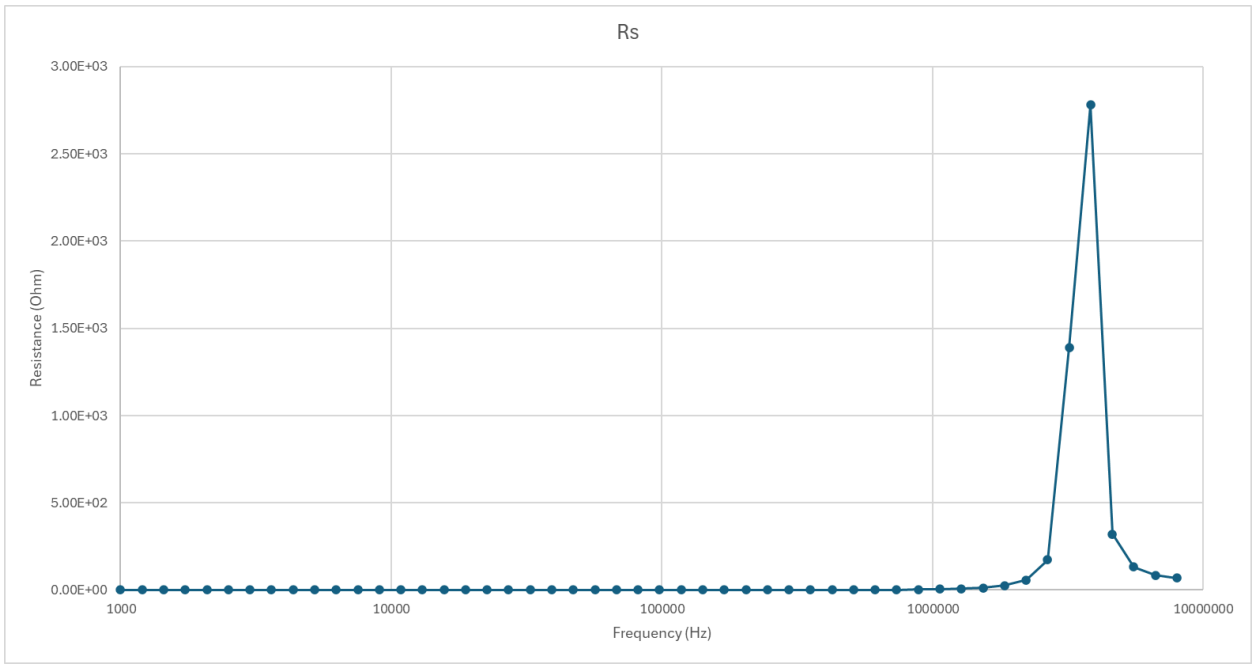
Procedure Conducted

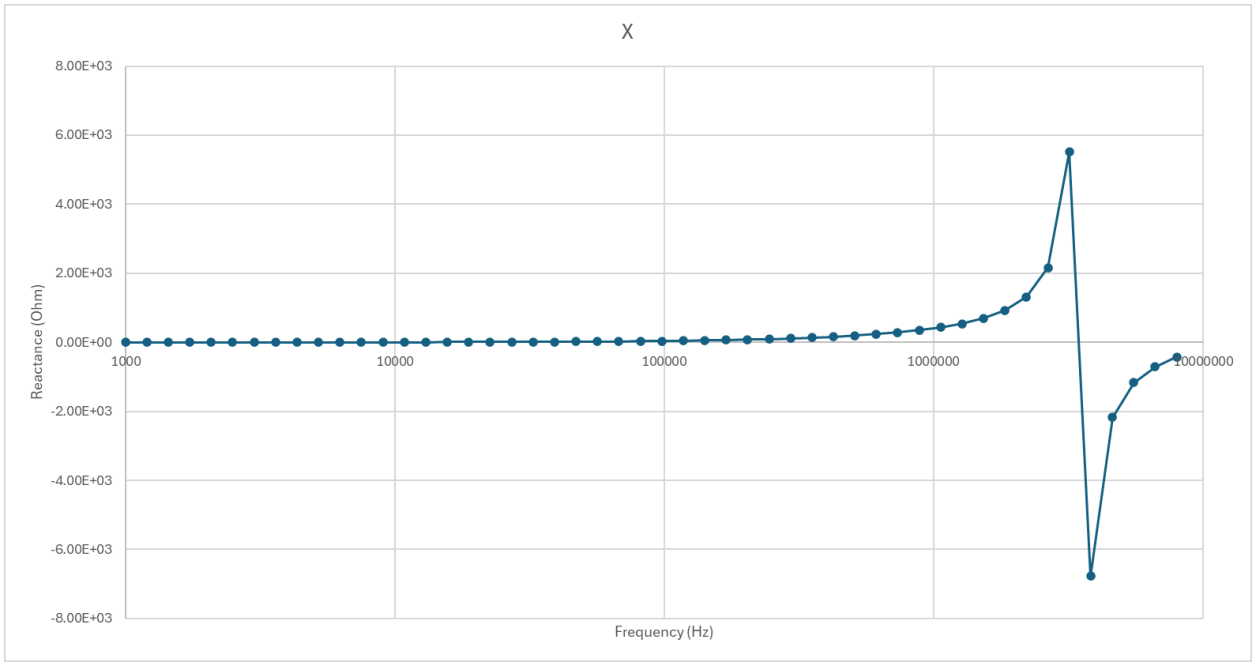
- Conducted a frequency sweep from 1 kHz to 8 MHz in 50 logarithmic steps, measuring the following parameters:
    - Coil impedance (Z)
    - Reactance (X)
    - Equivalent series resistance (Rs)
    - Equivalent series inductance (Ls)
    - Plotted the measured curves for analysis.
- Identified the self-resonant frequency of the inductor by refining the frequency sweep resolution or narrowing the sweep range around the resonance point.
- Assumed constant coil inductance and used measurement results to determine the equivalent model parameters of the inductor (referencing Figure 1.1).
- Conducted single-point testing at 100 kHz, measuring the inductor's impedance (Z), phase angle (θ), and equivalent series resistance (ESR).
- Calculated the coil's quality factor at 100 kHz and determined the basic and measurement accuracy of Z and θ using Section 10.6 of the LCR meter manual.
- Derived Zmin, Zmax, θmin, and θmax, then used these boundary values to determine the range of ΔESR.
- Evaluated the relative measurement error of ESR (%), including results, calculations, and explanations in the report.
- Assembled the resonant network on the ITR board and fine-tuned it to achieve a resonant frequency of 100 kHz.

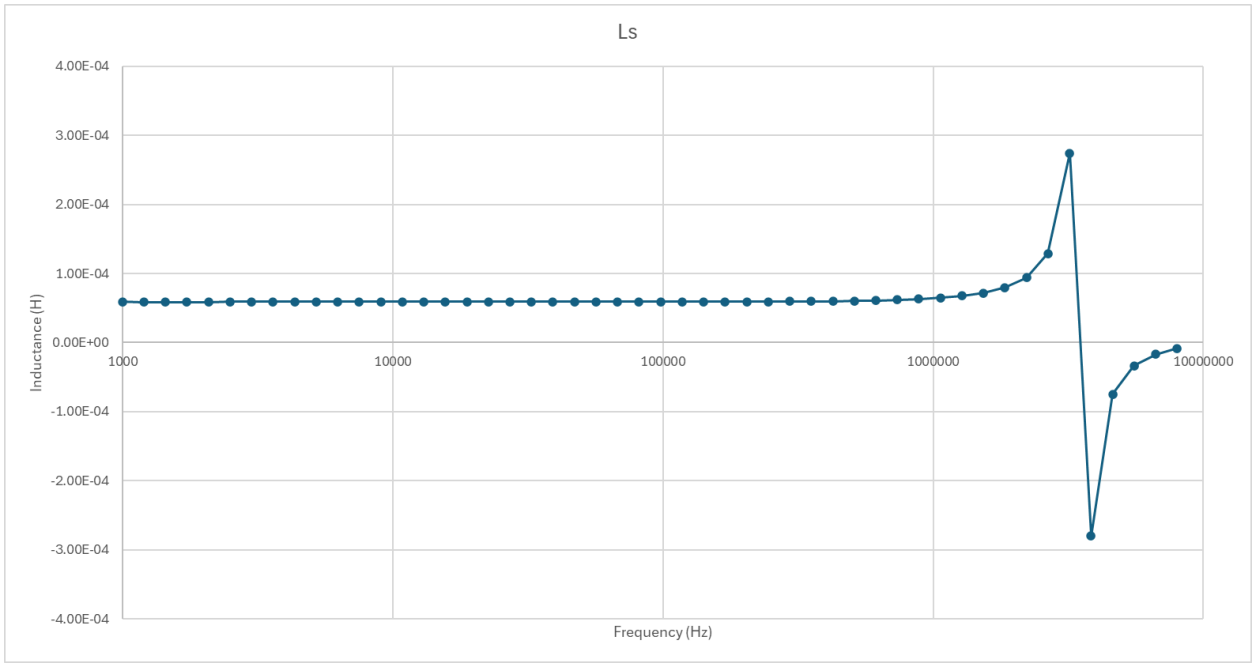
Observations and Results

In Task 1, a frequency sweep from **1 kHz to 8 MHz** was performed in **50 logarithmic steps**, and the results from the initial sweep are presented below.

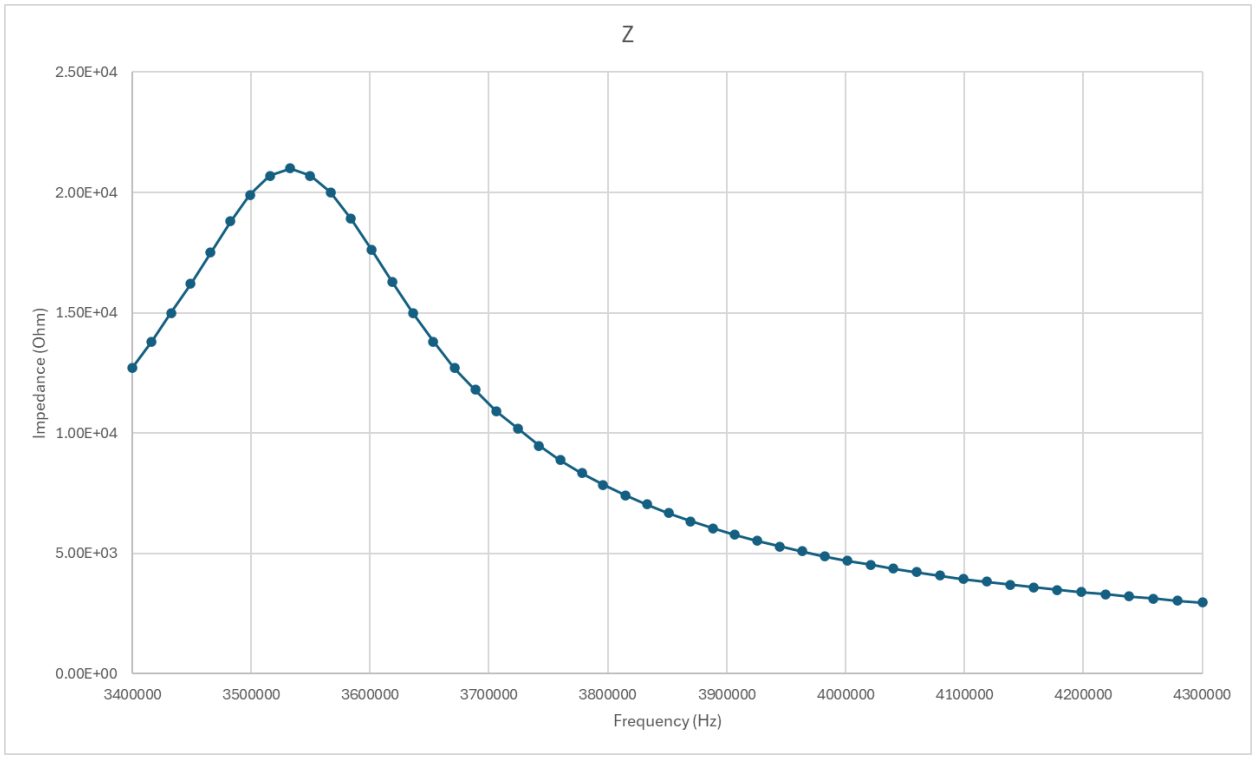


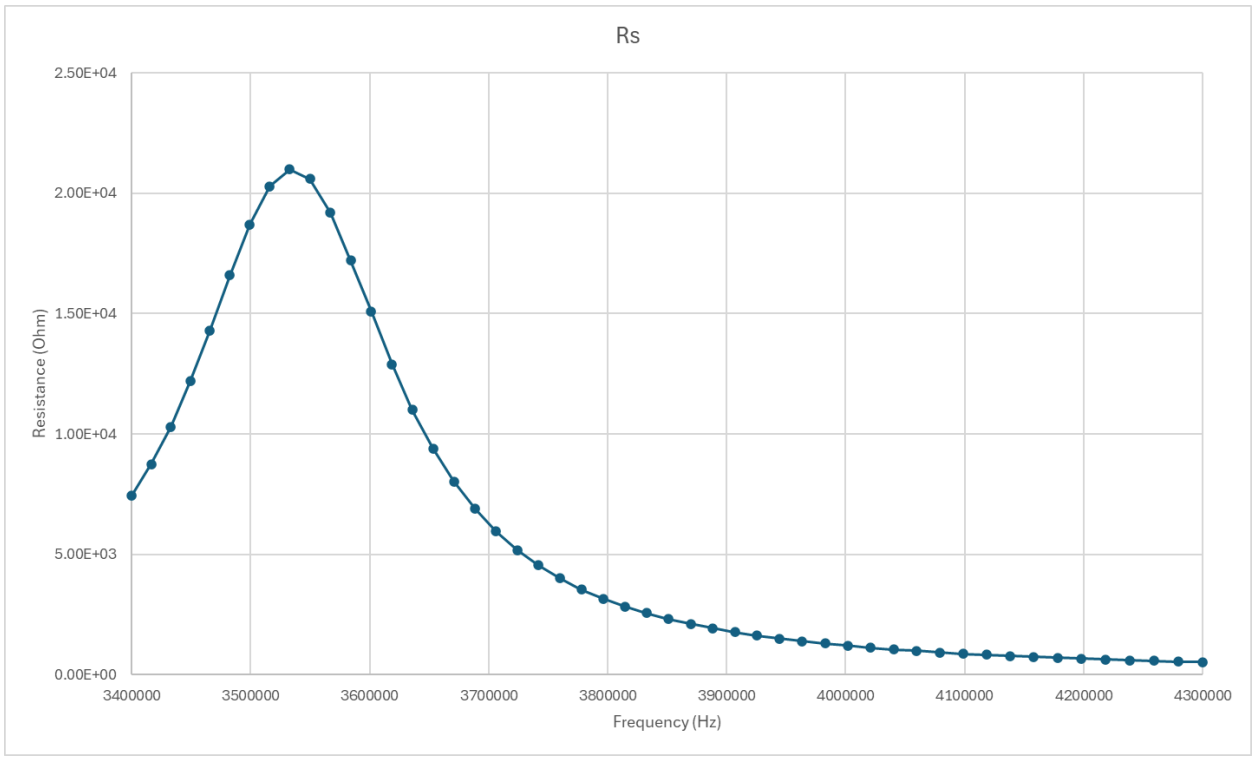


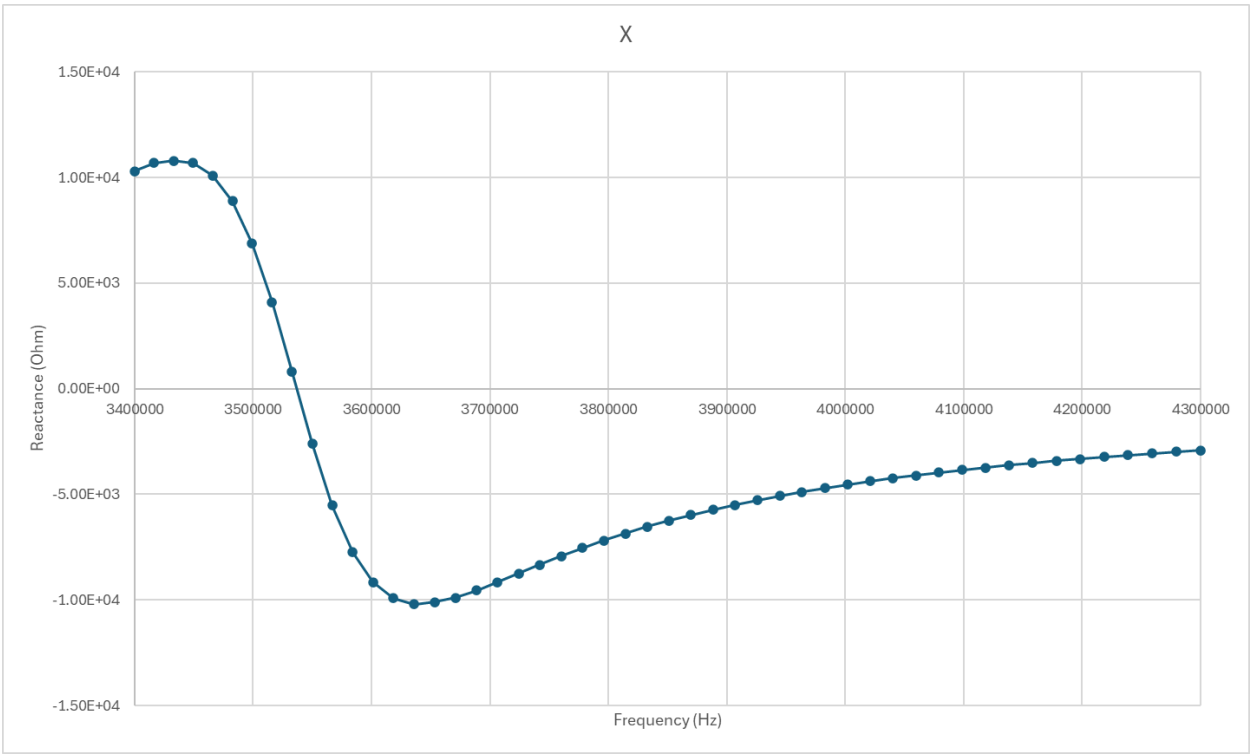

X
8.00E+03
6.00E+03
4.00E+03
2.00E+03
0.00E+00
-2.00E+03
-4.00E+03
-6.00E+03
-8.00E+03
Reactance (Ohm)
1000
10000
100000
1000000
10000000
Frequency (Hz)

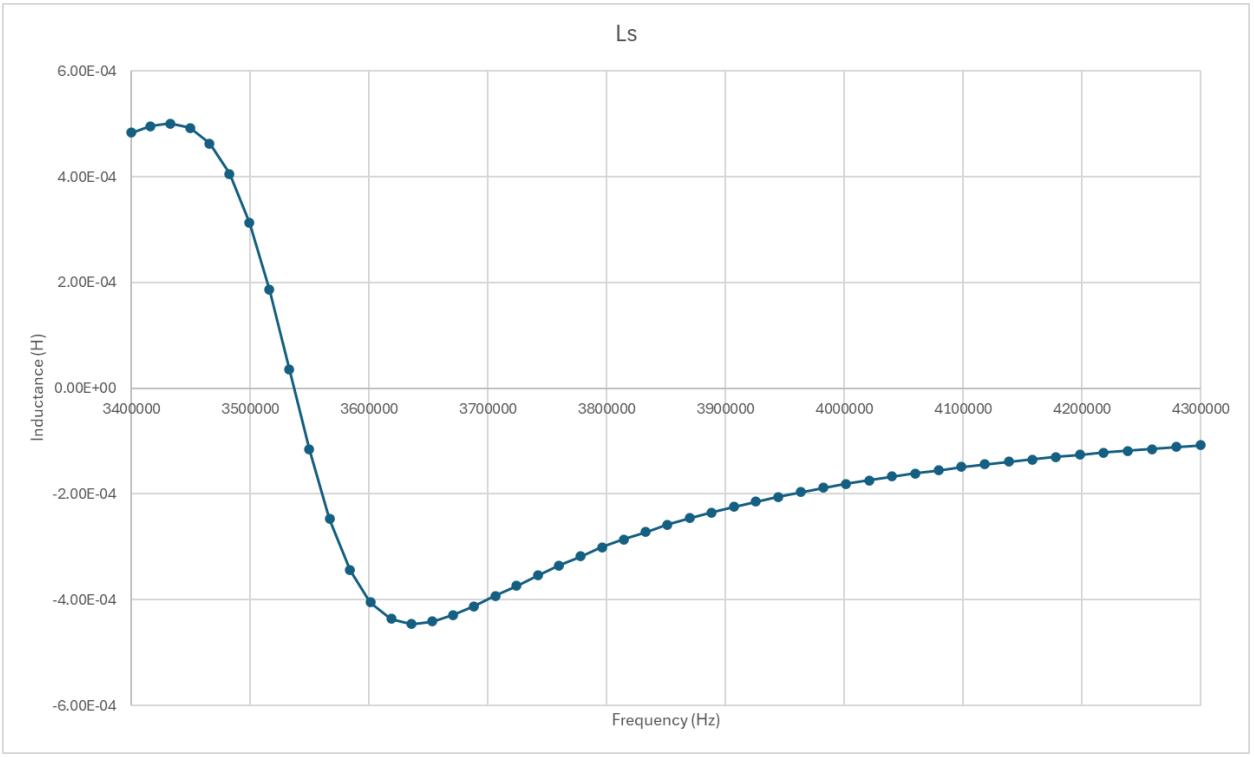

Ls
4.00E-04
3.00E-04
2.00E-04
1.00E-04
0.00E+00
-1.00E-04
-2.00E-04
-3.00E-04
-4.00E-04
Inductance (H)
1000
10000
100000
1000000
10000000
Frequency (Hz)

The initial frequency sweep results indicated that a more precise measurement of the self-resonant frequency required a narrower sweep range. To improve resolution, the range was adjusted to **3.4 MHz – 4.3 MHz**. The results from the second frequency sweep are presented below.

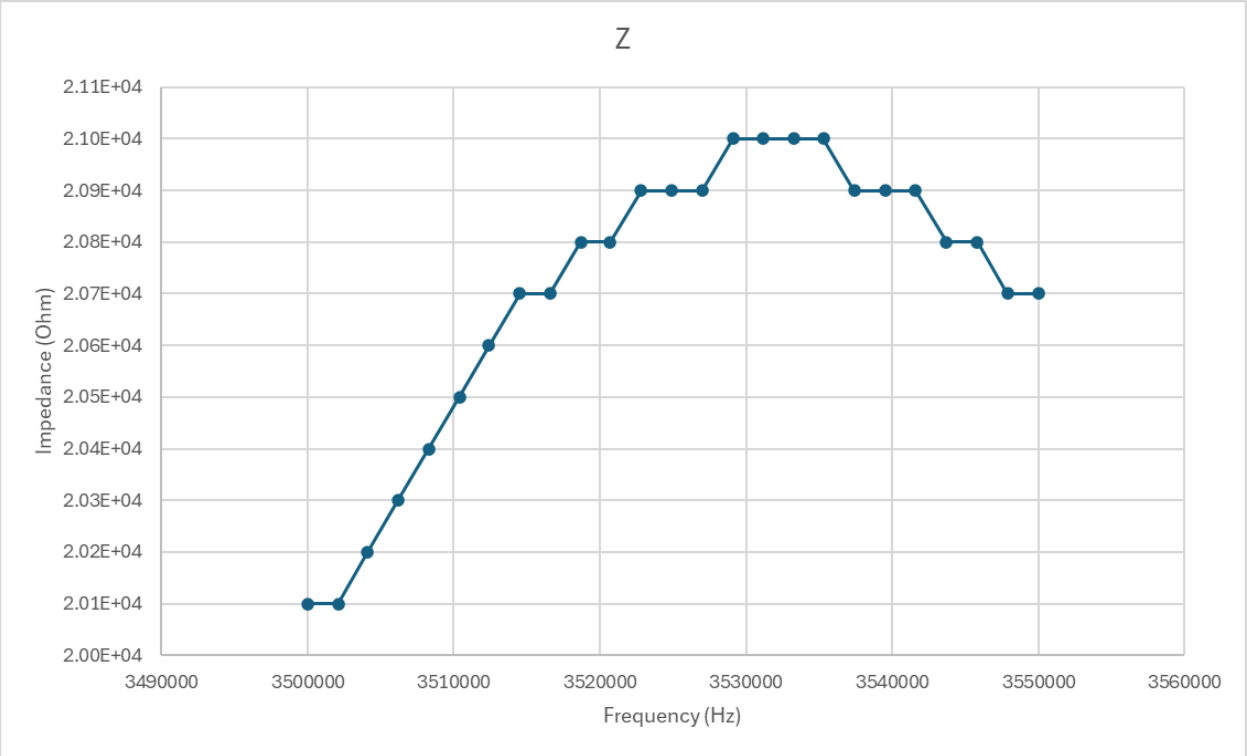


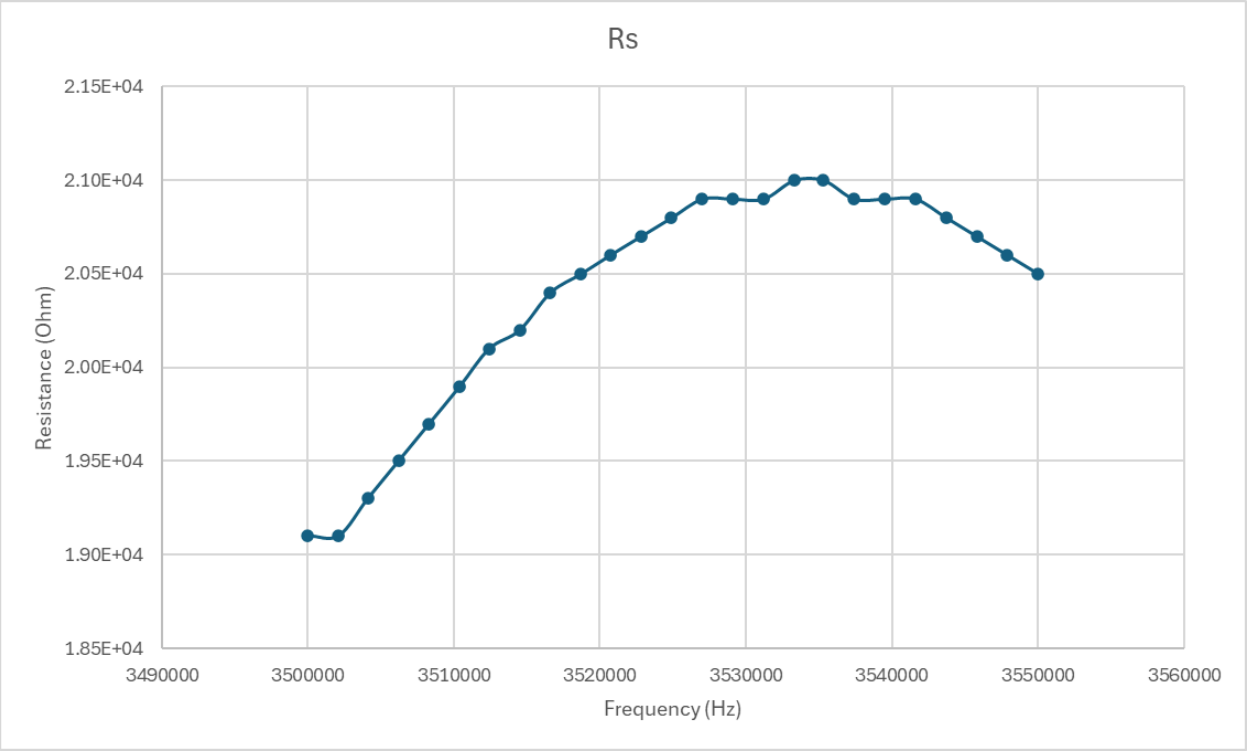


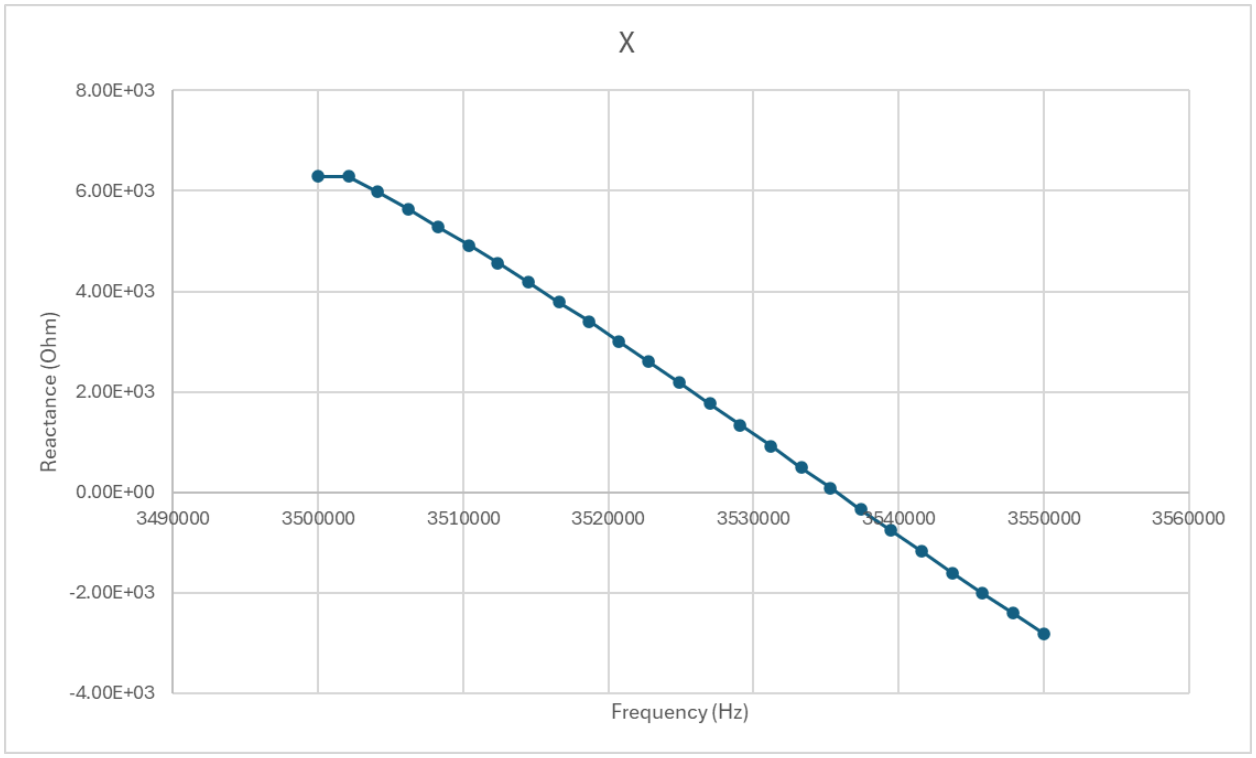

X
1.50E+04
1.00E+04
5.00E+03
0.00E+00
-5.00E+03
-1.00E+04
-1.50E+04
Reactance (Ohm)
3400000
3500000
3600000
3700000
3800000
3900000
4000000
4100000
4200000
4300000
Frequency (Hz)

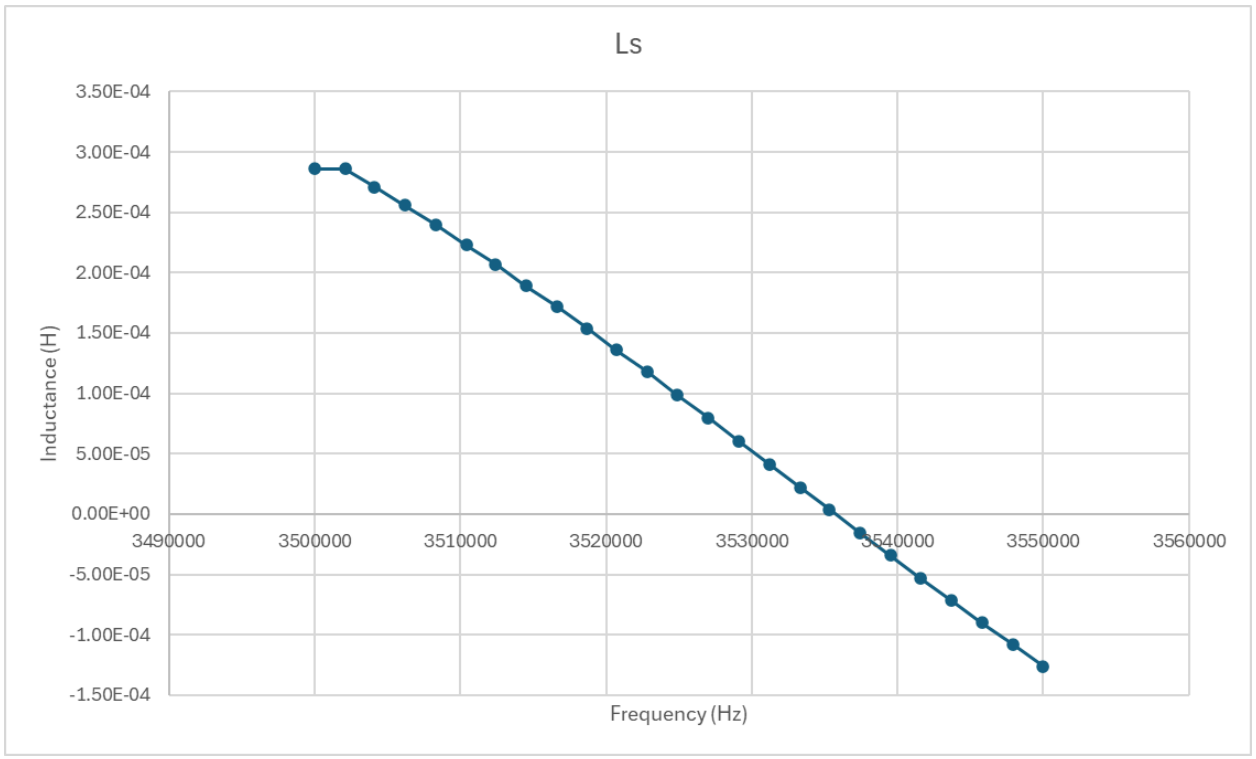

Ls
6.00E-04
4.00E-04
2.00E-04
0.00E+00
-2.00E-04
-4.00E-04
-6.00E-04
Inductance (H)
3400000
3500000
3600000
3700000
3800000
3900000
4000000
4100000
4200000
4300000
Frequency (Hz)

To obtain a more precise measurement of the self-resonant frequency, the process was repeated in **25 logarithmic steps** between the range of **3.5 MHz to 3.55 MHz**. The results from the third frequency sweep are presented below.









From the charts shown, the **self-resonant frequency** was determined to be **3535300 Hz.**

The equivalent model of the high-frequency model of the resonant inductor (referring to Fig. 1.1) uses the measurement of resonant inductance and self-resonant frequency to obtain the capacitance. The measured values are the following:

$L_r = 59.0761\ \mu H$

$f_{self-resonant} = 3535300\ Hz$

Since $wL = \frac{1}{wC}$ at self-resonant frequency,

The capacitance value is:

$$C_r = \frac{1}{(2\pi f)^2 \times L} = \frac{1}{(2\pi \times 3535300)^2 \times (59.0761E-6)} = 34.3064\ \rho F$$

Single-point testing was conducted at **100 kHz** to measure the inductor's impedance (Z), phase angle (θ), and equivalent series resistance (ESR). The measured values are the following:

- Z = 37.038 Ω
- θ = 89.90°
- ESR = 0.0612 Ω

Since the LCR meter **range** is set to be **1kΩ** (referring to Fig. 1.2), the basic accuracy equation is the following (referring to page 207 of the LCR Meter Manual):

$$Basic\ Accuracy = \pm\left(A + B \times \left|\frac{10 \times Zx}{Range} - 1\right|\right)$$

Range	For AC measurement (measurement frequency)					
	10.001 kHz to 100.00 kHz		100.01 kHz to 1 MHz		1.0001 MHz to 8 MHz	
100 MΩ	-	-	-	-	-	-
	-	-	-	-	-	-
10 MΩ	A=2	B=1	-	-	-	-
	A=2	B=1	-	-	-	-
1 MΩ	A=0.5	B=0.1	A=3	B=0.5	-	-
	A=0.6	B=0.1	A=3	B=0.5	-	-
100 kΩ	A=0.25	B=0.04	A=1	B=0.3	A=2	B=0.5
	A=0.2	B=0.02	A=1	B=0.3	A=2	B=0.3
10 kΩ	A=0.3	B=0.02	A=0.5	B=0.05	A=2	B=0.5
	A=0.2	B=0.02	A=0.5	B=0.05	A=1.5	B=0.3
1 kΩ	A=0.2	B=0.02	A=0.4	B=0.02	A=1.5	B=0.2
	A=0.15	B=0.02	A=0.4	B=0.02	A=1.5	B=0.2
100 Ω	A=0.2	B=0.02	A=0.5	B=0.03	A=1.5	B=0.2
	A=0.15	B=0.02	A=0.5	B=0.03	A=1.5	B=0.2
10 Ω	A=0.4	B=0.05	A=0.8	B=0.1	A=2	B=1.5
	A=0.3	B=0.03	A=0.5	B=0.05	A=2	B=1
1 Ω	A=1	B=0.3	A=1.5	B=1	A=3	B=3
	A=0.5	B=0.2	A=0.7	B=0.5	A=3	B=2
	A=2	B=2	A=4	B=3	-	-

The chart in the LCR meter manual was referenced to determine the values of A and B. Consequently, the basic accuracy equation was used for the calculations of $\mathbf{Z_{min}}$ and $\mathbf{Z_{max}}$ as follows:

$$Basic\ Accuracy\ of\ Z\ =\ \pm\left(0.2\ +\ 0.02\ \times\ \left|\frac{10\times 37.038}{1000}-1\right|\right)\ =\ 0.2125924\ \%$$

$$Z_{min}\ =\ 37.038\ \times\ \left(1-\frac{0.2126}{100}\right)\ =\ 36.9593\ \Omega$$

$$Z_{max}\ =\ 37.038\ \times\ \left(1+\frac{0.2126}{100}\right)\ =\ 37.1167\ \Omega$$

And for $\mathbf{\theta_{min}}$ and $\mathbf{\theta_{max}}$ calculations, the basic accuracy equation is the following:

$$Basic\ Accuracy\ of\ \theta\ =\ \pm\left(0.15\ +\ 0.02\ \times\ \left|\frac{10\times 37.038}{1000}-1\right|\right)\ =\ 0.1625924°$$

The error is observed to be quite large using given parameters.

The minimum and maximum values of the angle are,

$$\theta_{min}\ =\ 89.90\ -\ 0.1625924\ =\ 89.7374°$$

$$\theta_{max}\ =\ 89.90\ +\ 0.1625924\ =\ 90.0626°$$

It was observed that θ_{max} exceeded 90° after accounting for the basic accuracy margin. According to the lecture, inductance should never surpass 90°, as this would imply a negative resistance (real part), which is physically impossible. This value will be adjusted in the later calculations.

To find the inductance value for the quality factor of the coil. The below equation was used:

$$Z\ =\ R\ +\ jwL$$

$$|Z|\ =\ \sqrt{R^2\ +\ (wL)^2}$$

$$37.038\ =\ \sqrt{0.0612^2\ +\ (2\pi\ \times\ 100000\ \times\ L)^2}$$

$$L\ =\ 58.9477\ \mu H$$

The coil quality factor at 100 kHz is then calculated using the above inductance value,

$$Q\ =\ \frac{wL}{R}\ =\ \frac{(2\pi\times 100000\times 58.9477E-6)}{0.0612}\ =\ 605.1953$$

The boundary values of ΔESR are then determined using the following equations:

$$ESR_{min}\ =\ Z_{min}\,cos(\theta_{max})\ =\ 36.9593\ \times\ cos(90.0626)\ =\ -\ 0.0404\ \Omega$$

Since the ESR value cannot be negative, the benchmark ESR_{min} will equal 0.

$$ESR_{max} = Z_{max} cos(\theta_{min}) = 37.1167 \times cos(89.7374) = 0.1701\ \Omega$$

The relative measurement error of ESR in percentage is calculated as the following:

$$Error\ (\%) = \frac{0.1701 - 0.0612}{0.0612} \times 100 = 177.9411\ \%$$

The high error can be attributed to the 1 kΩ range setting of the LCR meter. The impedance (Z) consists of both resistance (R) and inductive reactance (ωL). While the resistance is in the milliohm range, the inductance measures approximately 37 Ω. This significant difference in magnitude likely contributed to the substantial ESR boundary error of 177% observed in the results.

Task 2: Resonant network assembling and tuning

Aim

In this part, the goal was to select the appropriate inductor and capacitors to best match the resonant tank tuning by testing on the LCR meter and choosing from the given list of components.

Test Setup and Procedure

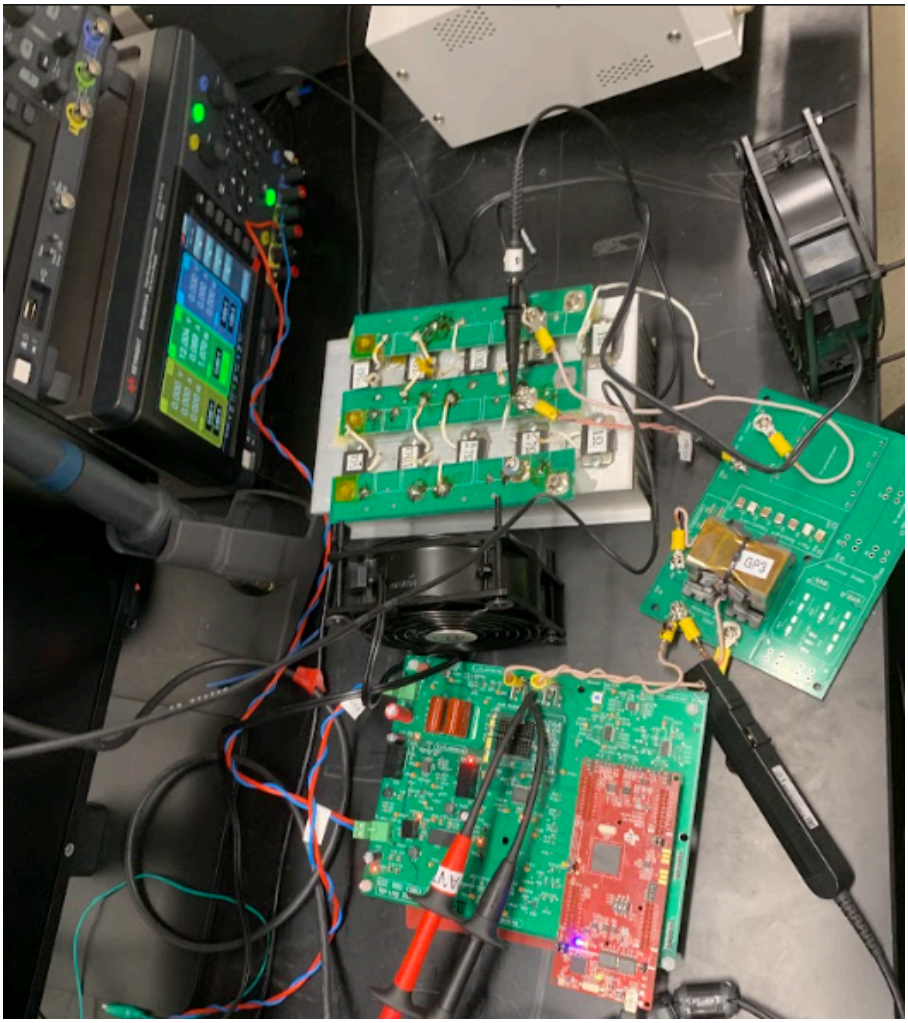


Fig 2.1: Setup for resonant tank matching after soldering components

1. The resonant network was assembled on the ITR board and tuned to 100 kHz. The inductor value of 59.07uH was used to select a suitable capacitor combination from 1 nF, 5.6 nF, and 6.8 nF.
2. The chosen capacitors of the combination 6.8×6+1×2, totaling 42.88nF, and the inductor were soldered onto the board.
3. An LCR meter was used to measure the reactance while sweeping the frequency to find the actual resonant point.

4. Observations were taken into csv files and the respective graphs were plotted to find the actual resonant point which was found to be around 104 kHz.
5. Adjustments were made as needed. The frequency was then swept from 75 kHz to 125 kHz in 51 steps to measure resistance and reactance.
6. The ESR was calculated and compared with previous results.

Observations and Results

Below data is the first set of logarithmic observations.

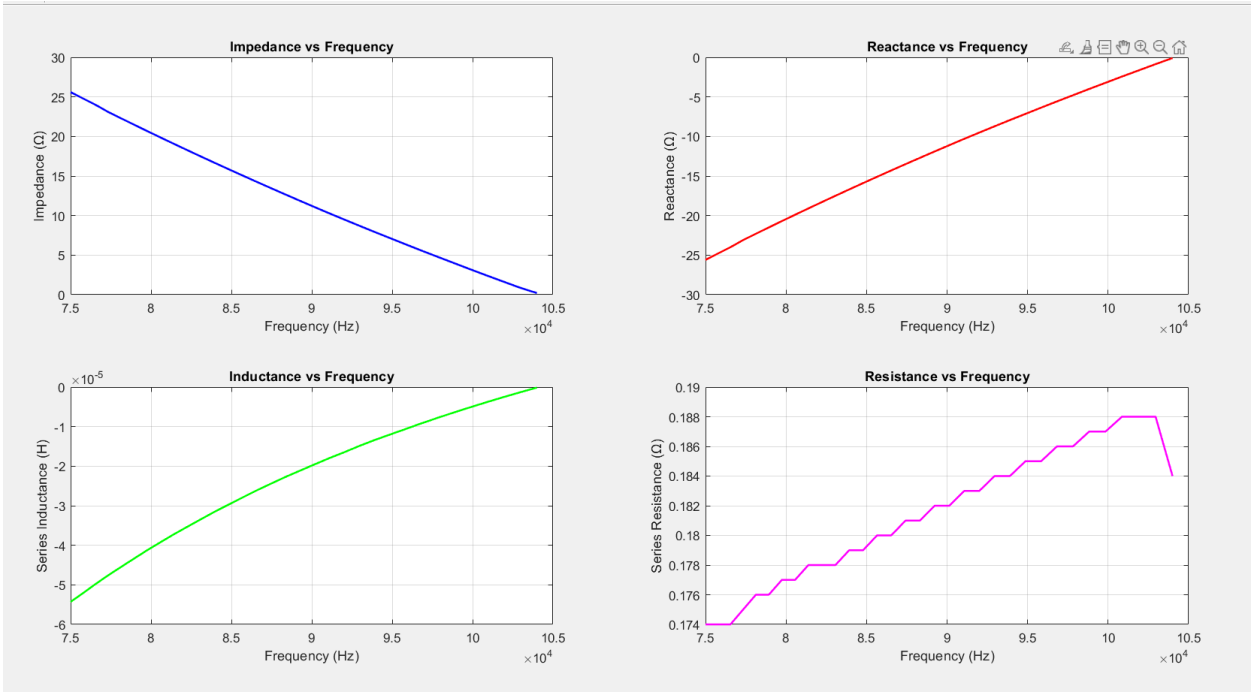
Frequency(Hz)	AC Status	Z	X	Ls	Rs
75000	2	2.56E+01	-2.56E+01	-5.43E-05	1.74E-01
75770	2	2.48E+01	-2.48E+01	-5.21E-05	1.74E-01
76548	2	2.40E+01	-2.40E+01	-4.98E-05	1.74E-01
77334	2	2.31E+01	-2.31E+01	-4.76E-05	1.75E-01
78128	2	2.23E+01	-2.23E+01	-4.55E-05	1.76E-01
78931	2	2.15E+01	-2.15E+01	-4.34E-05	1.76E-01
79741	2	2.07E+01	-2.07E+01	-4.13E-05	1.77E-01
80560	2	1.99E+01	-1.99E+01	-3.93E-05	1.77E-01
81387	2	1.91E+01	-1.91E+01	-3.73E-05	1.78E-01
82223	2	1.83E+01	-1.83E+01	-3.54E-05	1.78E-01

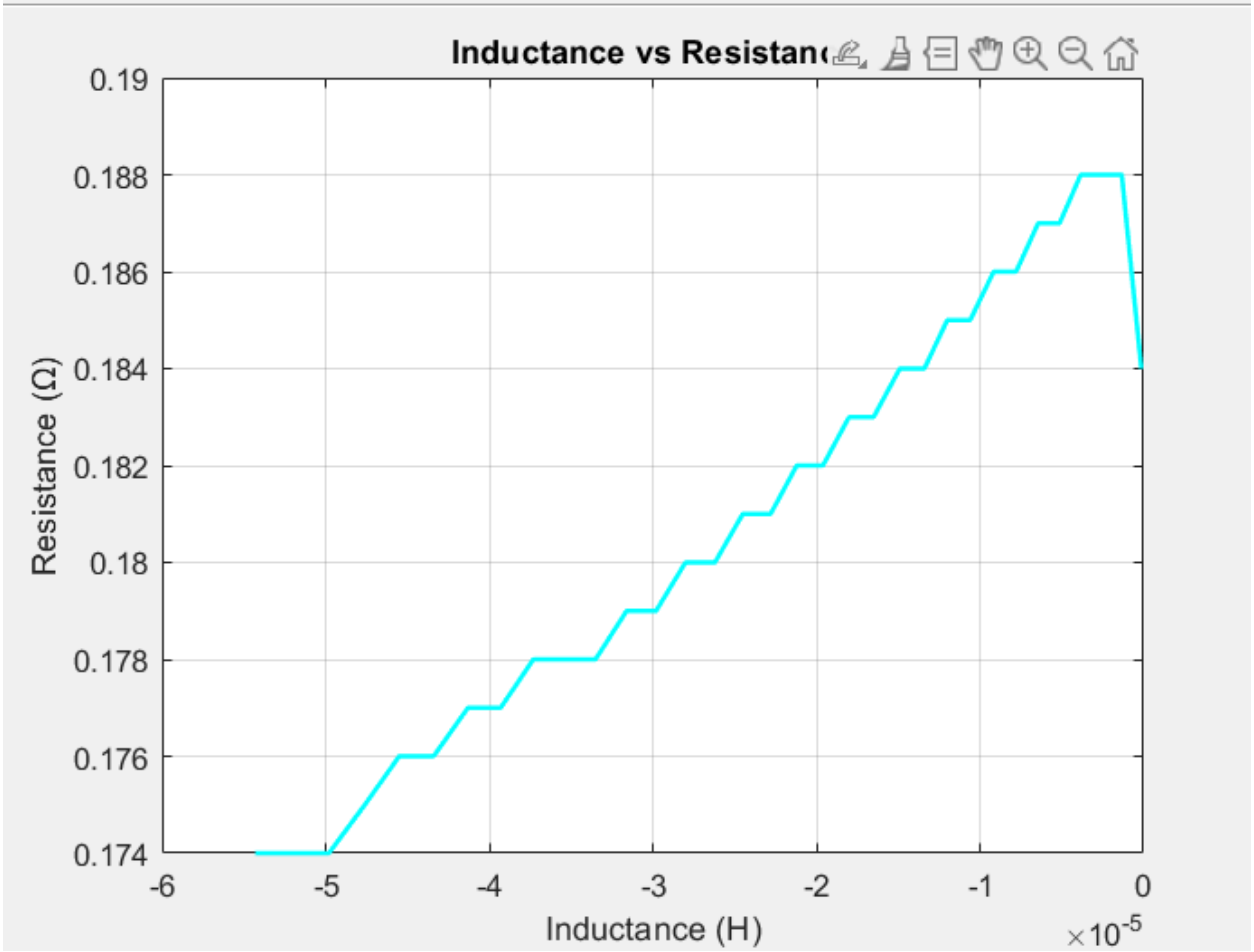
83067	2	1.75E+01	-1.75E+01	-3.35E-05	1.78E-01
83920	2	1.67E+01	-1.67E+01	-3.16E-05	1.79E-01
84782	2	1.59E+01	-1.59E+01	-2.98E-05	1.79E-01
85653	2	1.51E+01	-1.51E+01	-2.80E-05	1.80E-01
86532	2	1.43E+01	-1.43E+01	-2.62E-05	1.80E-01
87421	2	1.35E+01	-1.35E+01	-2.45E-05	1.81E-01
88319	2	1.27E+01	-1.27E+01	-2.28E-05	1.81E-01
89226	2	1.19E+01	-1.19E+01	-2.12E-05	1.82E-01
90142	2	1.11E+01	-1.11E+01	-1.96E-05	1.82E-01
91068	2	1.03E+01	-1.03E+01	-1.80E-05	1.83E-01
92003	2	9.51E+00	-9.51E+00	-1.65E-05	1.83E-01
92948	2	8.72E+00	-8.72E+00	-1.49E-05	1.84E-01
93902	2	7.93E+00	-7.93E+00	-1.34E-05	1.84E-01
94866	2	7.15E+00	-7.15E+00	-1.20E-05	1.85E-01

95840	2	6.36E+00	-6.36E+00	-1.06E-05	1.85E-01
96825	2	5.57E+00	-5.57E+00	-9.16E-06	1.86E-01
97819	2	4.79E+00	-4.79E+00	-7.79E-06	1.86E-01
98823	2	4.01E+00	-4.00E+00	-6.44E-06	1.87E-01
99838	2	3.22E+00	-3.22E+00	-5.13E-06	1.87E-01
100860	2	2.44E+00	-2.43E+00	-3.84E-06	1.88E-01
101900	2	1.66E+00	-1.65E+00	-2.57E-06	1.88E-01
102950	2	8.79E-01	-8.59E-01	-1.33E-06	1.88E-01
104000	2	2.04E-01	-8.81E-02	-1.35E-07	1.84E-01
105070	2	7.18E-01	6.92E-01	1.05E-06	1.90E-01
106150	2	1.49E+00	1.48E+00	2.21E-06	1.91E-01
107240	2	2.27E+00	2.26E+00	3.36E-06	1.92E-01
108340	2	3.05E+00	3.05E+00	4.47E-06	1.93E-01
109450	2	3.83E+00	3.83E+00	5.57E-06	1.93E-01

110580	2	4.62E+00	4.62E+00	6.65E-06	1.94E-01
111710	2	5.40E+00	5.40E+00	7.69E-06	1.95E-01
112860	2	6.19E+00	6.19E+00	8.73E-06	1.95E-01
114020	2	6.98E+00	6.98E+00	9.74E-06	1.96E-01
115190	2	7.77E+00	7.76E+00	1.07E-05	1.97E-01
116370	2	8.55E+00	8.55E+00	1.17E-05	1.97E-01
117570	2	9.35E+00	9.34E+00	1.26E-05	1.98E-01
118780	2	1.01E+01	1.01E+01	1.36E-05	1.99E-01
119990	2	1.09E+01	1.09E+01	1.45E-05	1.99E-01
121230	2	1.17E+01	1.17E+01	1.54E-05	2.00E-01
122470	2	1.25E+01	1.25E+01	1.63E-05	2.01E-01
123730	2	1.33E+01	1.33E+01	1.71E-05	2.01E-01
125000	2	1.41E+01	1.41E+01	1.80E-05	2.02E-01

The following graphical results were yielded from the observations:





The observations were taken in a linearly varying fashion and yielded the following graphical results.

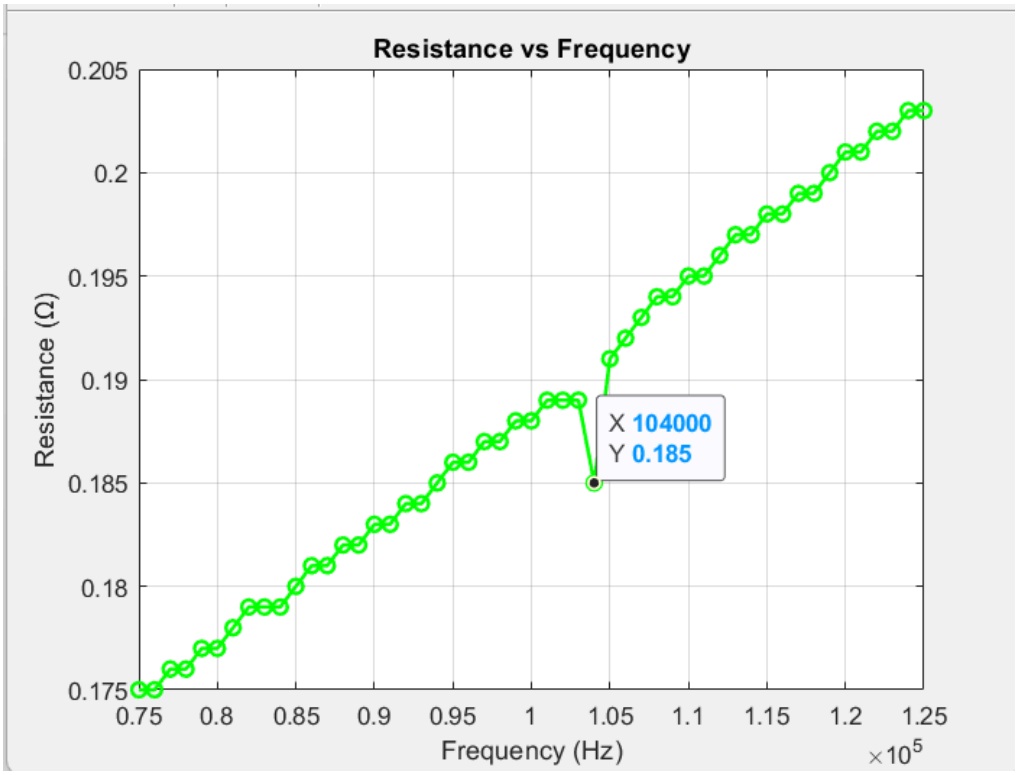
Frequency(Hz)	AC Status	Z	X	Ls	Rs
75000	2	2.56E+01	-2.56E+01	-5.44E-05	1.75E-01
76000	2	2.46E+01	-2.45E+01	-5.14E-05	1.75E-01
77000	2	2.35E+01	-2.35E+01	-4.86E-05	1.76E-01
78000	2	2.25E+01	-2.25E+01	-4.58E-05	1.76E-01
79000	2	2.15E+01	-2.14E+01	-4.32E-05	1.77E-01
80000	2	2.05E+01	-2.04E+01	-4.07E-05	1.77E-01
81000	2	1.95E+01	-1.95E+01	-3.82E-05	1.78E-01
82000	2	1.85E+01	-1.85E+01	-3.59E-05	1.79E-01
83000	2	1.75E+01	-1.75E+01	-3.36E-05	1.79E-01
84000	2	1.66E+01	-1.66E+01	-3.14E-05	1.79E-01
85000	2	1.57E+01	-1.57E+01	-2.93E-05	1.80E-01
86000	2	1.48E+01	-1.48E+01	-2.73E-05	1.81E-01
87000	2	1.39E+01	-1.39E+01	-2.53E-05	1.81E-01

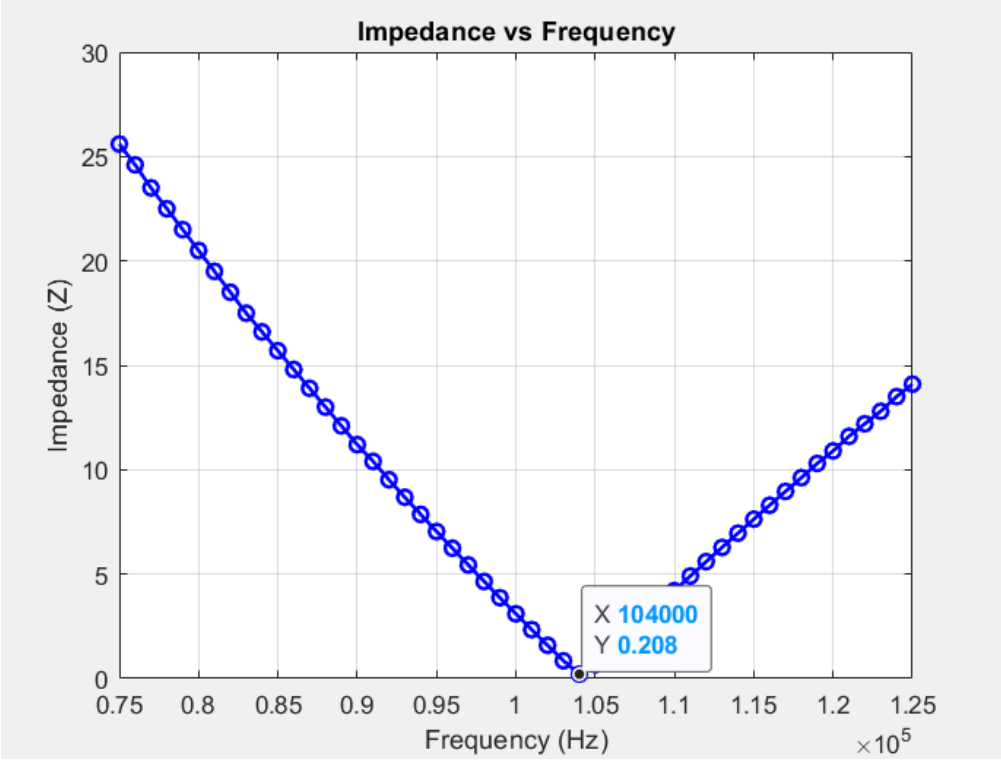
88000	2	1.30E+01	-1.30E+01	-2.34E-05	1.82E-01
89000	2	1.21E+01	-1.21E+01	-2.16E-05	1.82E-01
90000	2	1.12E+01	-1.12E+01	-1.98E-05	1.83E-01
91000	2	1.04E+01	-1.04E+01	-1.81E-05	1.83E-01
92000	2	9.52E+00	-9.52E+00	-1.65E-05	1.84E-01
93000	2	8.68E+00	-8.68E+00	-1.49E-05	1.84E-01
94000	2	7.86E+00	-7.86E+00	-1.33E-05	1.85E-01
95000	2	7.04E+00	-7.04E+00	-1.18E-05	1.86E-01
96000	2	6.24E+00	-6.24E+00	-1.03E-05	1.86E-01
97000	2	5.44E+00	-5.44E+00	-8.92E-06	1.87E-01
98000	2	4.65E+00	-4.65E+00	-7.55E-06	1.87E-01
99000	2	3.87E+00	-3.87E+00	-6.22E-06	1.88E-01
100000	2	3.10E+00	-3.10E+00	-4.93E-06	1.88E-01
101000	2	2.34E+00	-2.33E+00	-3.67E-06	1.89E-01

102000	2	1.59E+00	-1.58E+00	-2.46E-06	1.89E-01
103000	2	8.49E-01	-8.27E-01	-1.28E-06	1.89E-01
104000	2	2.08E-01	-9.38E-02	-1.44E-07	1.85E-01
105000	2	6.63E-01	6.35E-01	9.63E-07	1.91E-01
106000	2	1.38E+00	1.36E+00	2.05E-06	1.92E-01
107000	2	2.09E+00	2.08E+00	3.10E-06	1.93E-01
108000	2	2.80E+00	2.80E+00	4.12E-06	1.94E-01
109000	2	3.51E+00	3.51E+00	5.12E-06	1.94E-01
110000	2	4.21E+00	4.21E+00	6.09E-06	1.95E-01
111000	2	4.91E+00	4.90E+00	7.03E-06	1.95E-01
112000	2	5.60E+00	5.59E+00	7.95E-06	1.96E-01
113000	2	6.28E+00	6.28E+00	8.84E-06	1.97E-01
114000	2	6.96E+00	6.96E+00	9.71E-06	1.97E-01
115000	2	7.63E+00	7.63E+00	1.06E-05	1.98E-01

116000	2	8.30E+00	8.30E+00	1.14E-05	1.98E-01
117000	2	8.96E+00	8.96E+00	1.22E-05	1.99E-01
118000	2	9.62E+00	9.62E+00	1.30E-05	1.99E-01
119000	2	1.03E+01	1.03E+01	1.37E-05	2.00E-01
120000	2	1.09E+01	1.09E+01	1.45E-05	2.01E-01
121000	2	1.16E+01	1.16E+01	1.52E-05	2.01E-01
122000	2	1.22E+01	1.22E+01	1.59E-05	2.02E-01
123000	2	1.28E+01	1.28E+01	1.66E-05	2.02E-01
124000	2	1.35E+01	1.35E+01	1.73E-05	2.03E-01
125000	2	1.41E+01	1.41E+01	1.80E-05	2.03E-01

The following graphical results were yielded from the observation.





The plotted graphs illustrate the frequency response of the assembled resonant tank circuit. Resistance increases steadily with frequency, while impedance exhibits a distinct dip at approximately **104 kHz**, indicating the actual resonance point. At this frequency, inductive and capacitive reactances cancel each other, minimizing impedance and leaving only the resistive component. This confirms the successful tuning of the resonant network, aligning closely with the **expected 100 kHz target**. The smooth resistance trend suggests it has minimal impact on resonance behavior, reinforcing the effectiveness of the selected inductor and capacitor combination.

To find out the error percentage, we use the same methodology as in Task 1, with a different range:

Range	For AC measurement (measurement frequency)					
	10.001 kHz to 100.00 kHz		100.01 kHz to 1 MHz		1.0001 MHz to 8 MHz	
100 MΩ	-	-	-	-	-	-
	-	-	-	-	-	-
10 MΩ	A=2	B=1	-	-	-	-
	A=2	B=1	-	-	-	-
1 MΩ	A=0.5	B=0.1	A=3	B=0.5	-	-
	A=0.6	B=0.1	A=3	B=0.5	-	-
100 kΩ	A=0.25	B=0.04	A=1	B=0.3	A=2	B=0.5
	A=0.2	B=0.02	A=1	B=0.3	A=2	B=0.3
10 kΩ	A=0.3	B=0.02	A=0.5	B=0.05	A=2	B=0.5
	A=0.2	B=0.02	A=0.5	B=0.05	A=1.5	B=0.3
1 kΩ	A=0.2	B=0.02	A=0.4	B=0.02	A=1.5	B=0.2
	A=0.15	B=0.02	A=0.4	B=0.02	A=1.5	B=0.2
100 Ω	A=0.2	B=0.02	A=0.5	B=0.03	A=1.5	B=0.2
	A=0.15	B=0.02	A=0.5	B=0.03	A=1.5	B=0.2
10 Ω	A=0.4	B=0.05	A=0.8	B=0.1	A=2	B=1.5
	A=0.3	B=0.03	A=0.5	B=0.05	A=2	B=1
1 Ω	A=1	B=0.3	A=1.5	B=1	A=3	B=3
	A=0.5	B=0.2	A=0.7	B=0.5	A=3	B=2
100 mΩ	A=2	B=2	A=4	B=3	-	-
	A=2	B=1.5	A=3	B=4	-	-

$$Basic\ Accuracy\ of\ Z = \pm\left(0.4 + 0.02 \times \left|\frac{10 \times 0.208}{1000} - 1\right|\right) = 0.38\%$$

$$Z_{min} = 0.208 \times \left(1 - \frac{0.38}{100}\right) = 0.2072\ \Omega$$

$$Z_{max} = 0.208 \times \left(1 + \frac{0.38}{100}\right) = 0.2088\ \Omega$$

And for $\boldsymbol{\theta_{min}}$ and $\boldsymbol{\theta_{max}}$ calculations, the basic accuracy equation is the following:

$$Basic\ Accuracy\ of\ \theta\ =\ \pm\left(0.4\ +\ 0.02\ \times\ \left|\frac{10\times 0.208}{1000}-1\right|\right)\ =\ 0.38°$$

The minimum and maximum values of the angle are,

$$\theta_{min}\ =\ 27\ -\ 0.38\ =\ 26.62°$$

$$\theta_{max}\ =\ 27\ +\ 0.38\ =\ 27.38°$$

The boundary values of ΔESR are then determined using the following equations:

$$ESR_{min}\ =\ Z_{min}\ cos(\theta_{max})\ =\ 0.2072\ \times\ cos(27.38)\ =\ 0.1840\ \Omega$$

$$ESR_{max}\ =\ Z_{max}\ cos(\theta_{min})\ =\ 0.2088\ \times\ cos(26.62)\ =\ 0.1867\ \Omega$$

The relative measurement error of ESR in percentage is calculated as the following:

$$Error\ (\%)\ =\frac{0.1867-0.1840}{0.1840}\times\ 100\ =\ 1.4674\%$$

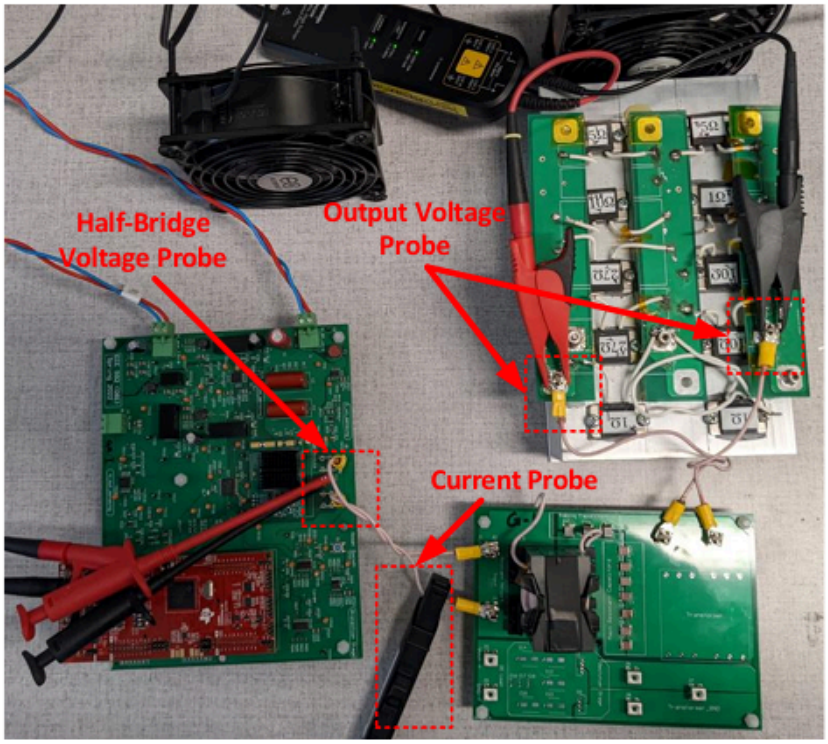
With precise tuning of the resonant components, the error percentage drops significantly. This can be attributed to the low impedance and reactance values at precisely the point of the resonance frequency.

Task 3: Measuring the inverter's DC-AC voltage gain and efficiency

Aim

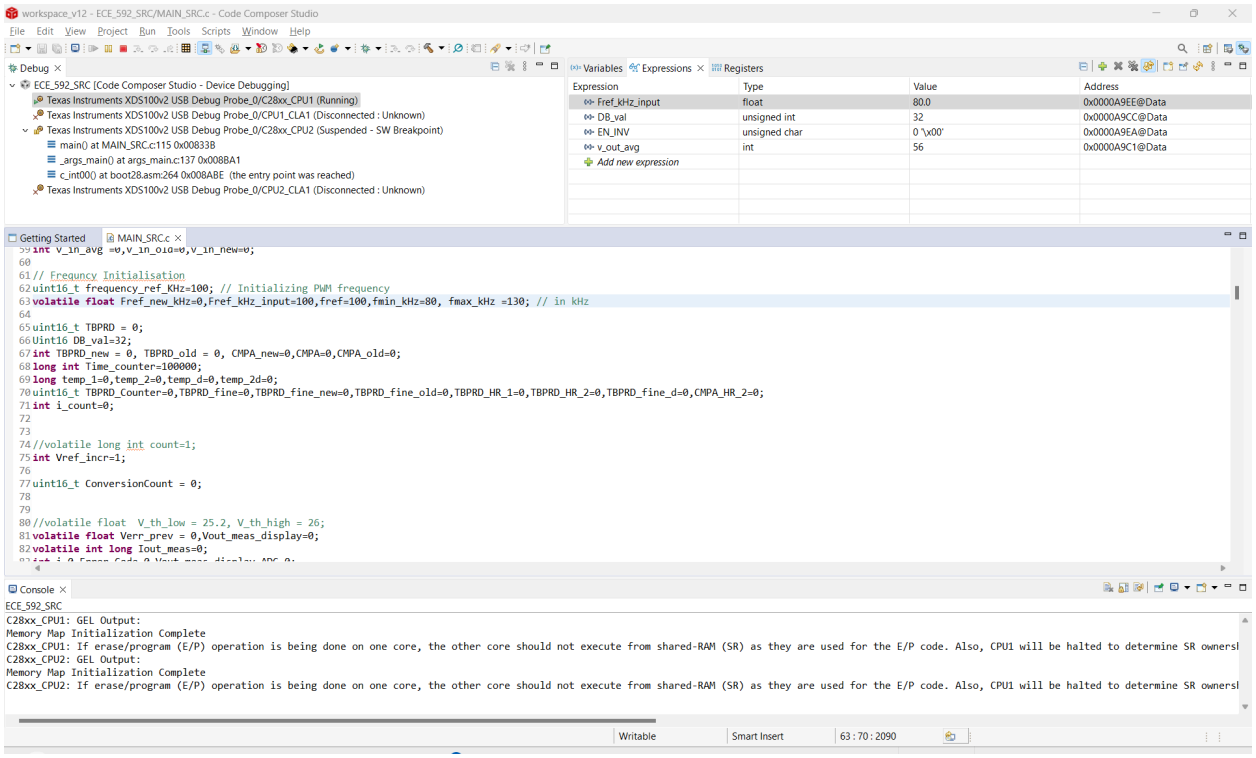
In this task, the inverter's DC-AC gain function and efficiency will be measured for different combinations of the input DC voltage and the load. The experiment will use the main power supply and one auxiliary power supply to power the switching board and the resistive bank to load the inverter.

Test Setup



Procedure Conducted

- The resistor banks were soldered to the required value of 10 ohms and 8 ohms.
- A resonance point was found with the resistor bank connected to the ITR board and sweeping the frequency from 95 kHz to 105 kHz.
- Add the frequency variable 'Fref_kHz_input' into the 'Expression' section of the CCS Debug window. Activate the continuous refresh to track the actual value of variables in the Debug window. Set the frequency to 75 kHz. We tried setting the Fref_kHz_input to 75 but the output was **80** (kHz).



- Frequency was changed in 5-kHz steps between 80 kHz to 95 kHz, 1-kHz steps between 95 kHz to 105 kHz, and 5-kHz steps between 105 kHz to 125 kHz. Each time, the output voltage and power values from the oscilloscope and the input power displayed on the main power supply were noted to calculate the efficiency and the DC-AC voltage gain.

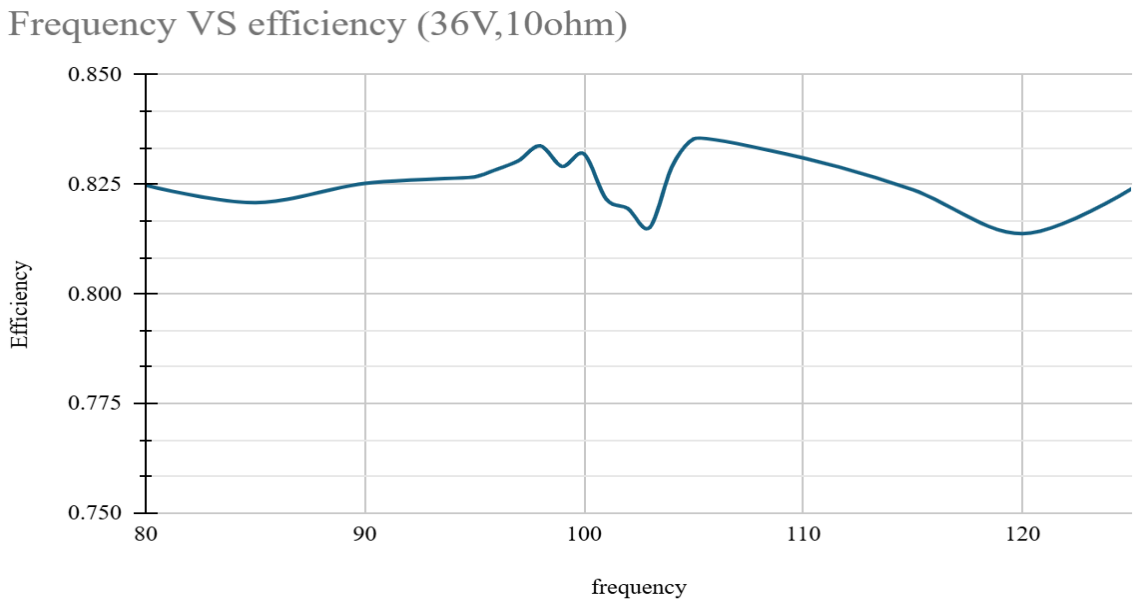
Observations and Results

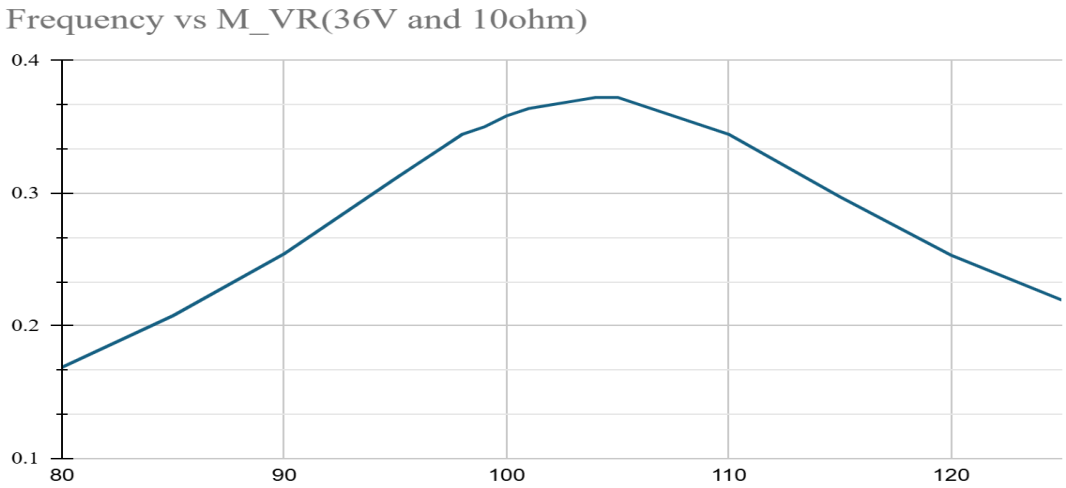
Ri (targeted)	Ri (measured)	Li(measured)
10Ω	10.033Ω	0.32μH
8Ω	8.1844	0.45μH

Below is the table and graph of different combinations of voltage and resistance:

(36V,10Ω)

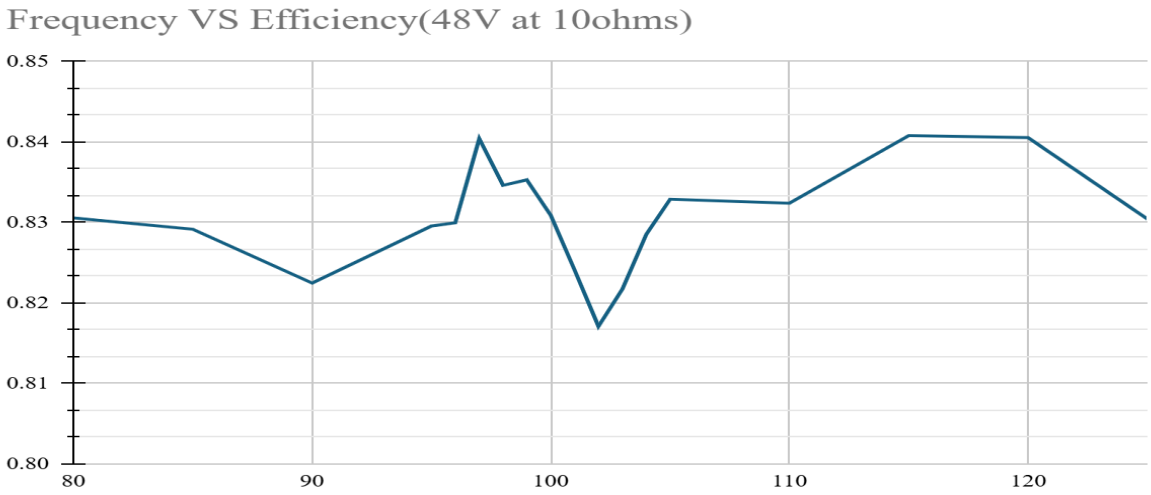
Input current	Voltage	Current	Frequency	Win	Wout	efficiency	M_VR
0.146	6.08	0.713	80	5.256	4.33504	0.8247792998	0.1688888889
0.222	7.48	0.877	85	7.992	6.55996	0.8208158158	0.2077777778
0.333	9.16	1.08	90	11.988	9.8928	0.8252252252	0.2544444444
0.493	11.2	1.31	95	17.748	14.672	0.8266846969	0.3111111111
0.529	11.6	1.36	96	19.044	15.776	0.8283973955	0.3222222222
0.566	12	1.41	97	20.376	16.92	0.8303886926	0.3333333333
0.599	12.4	1.45	98	21.564	17.98	0.8337970692	0.3444444444
0.629	12.6	1.49	99	22.644	18.774	0.8290937997	0.35
0.659	12.9	1.53	100	23.724	19.737	0.8319423369	0.3583333333
0.682	13.1	1.54	101	24.552	20.174	0.8216845878	0.3638888889
0.698	13.2	1.56	102	25.128	20.592	0.8194842407	0.3666666667
0.707	13.3	1.56	103	25.452	20.748	0.8151815182	0.3694444444
0.705	13.4	1.57	104	25.38	21.038	0.8289204098	0.3722222222
0.704	13.4	1.58	105	25.344	21.172	0.835385101	0.3722222222
0.601	12.4	1.45	110	21.636	17.98	0.8310223701	0.3444444444
0.451	10.7	1.25	115	16.236	13.375	0.823786647	0.2972222222
0.33	9.12	1.06	120	11.88	9.6672	0.8137373737	0.2533333333
0.245	7.9	0.92	125	8.82	7.268	0.8240362812	0.2194444444

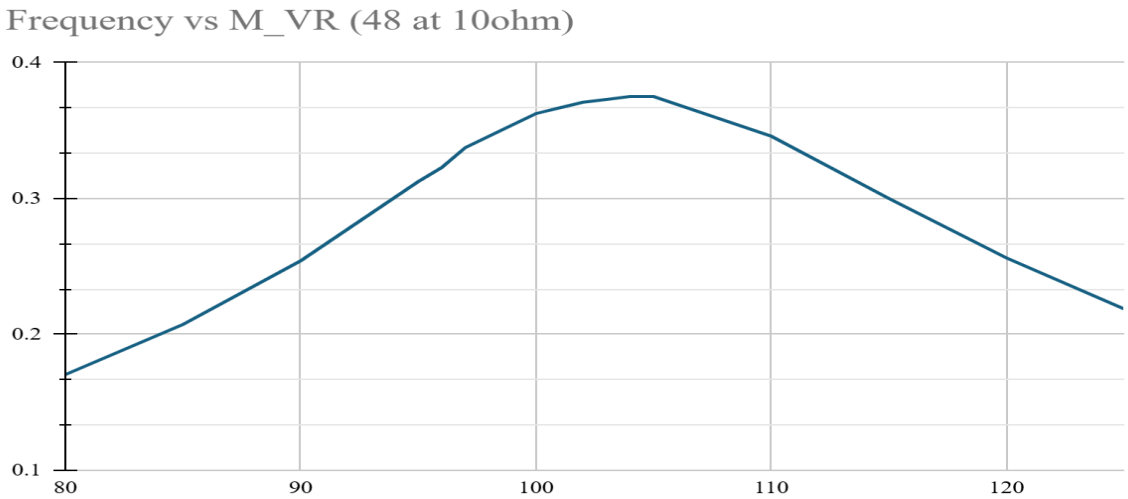




(48V,10Ω)

Input current	Voltage	Current	Frequency	Win	Wout	efficiency	M_VR
0.198	8.18	0.965	80	9.504	7.8937	0.8305660774	0.1704166667
0.295	9.95	1.18	85	14.16	11.741	0.8291666667	0.2072916667
0.445	12.2	1.44	90	21.36	17.568	0.8224719101	0.2541666667
0.663	15	1.76	95	31.824	26.4	0.8295625943	0.3125
0.712	15.5	1.83	96	34.176	28.365	0.8299683989	0.3229166667
0.763	16.2	1.9	97	36.624	30.78	0.8404325033	0.3375
0.808	16.6	1.95	98	38.784	32.37	0.8346225248	0.3458333333
0.848	17	2	99	40.704	34	0.8352987421	0.3541666667
0.89	17.4	2.04	100	42.72	35.496	0.8308988764	0.3625
0.921	17.6	2.07	101	44.208	36.432	0.8241042345	0.3666666667
0.944	17.8	2.08	102	45.312	37.024	0.8170903955	0.3708333333
0.953	17.9	2.1	103	45.744	37.59	0.8217471144	0.3729166667
0.955	18	2.11	104	45.84	37.98	0.8285340314	0.375
0.95	18	2.11	105	45.6	37.98	0.8328947368	0.375
0.806	16.6	1.94	110	38.688	32.204	0.8324028122	0.3458333333
0.603	14.4	1.69	115	28.944	24.336	0.8407960199	0.3
0.439	12.3	1.44	120	21.072	17.712	0.840546697	0.25625
0.324	10.5	1.23	125	15.552	12.915	0.8304398148	0.21875

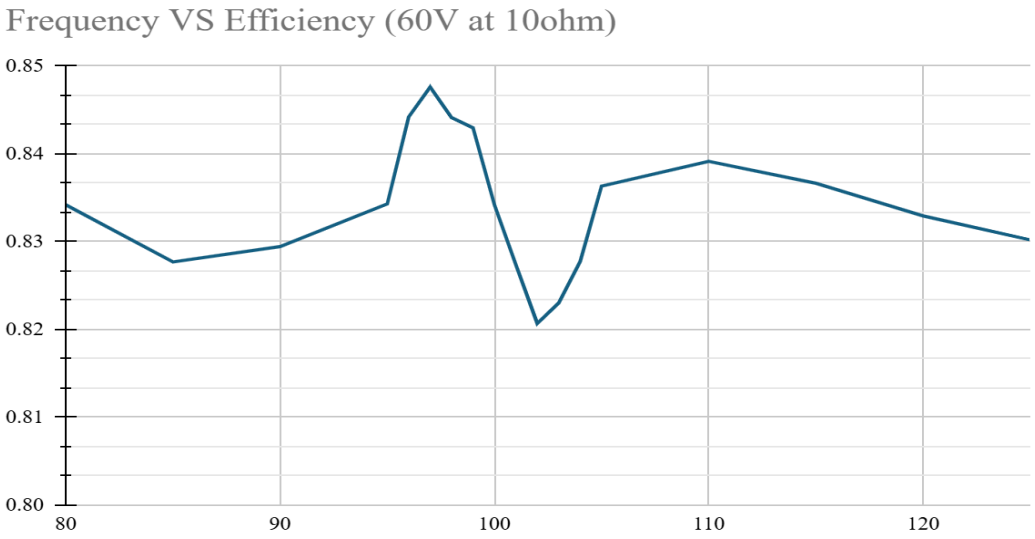


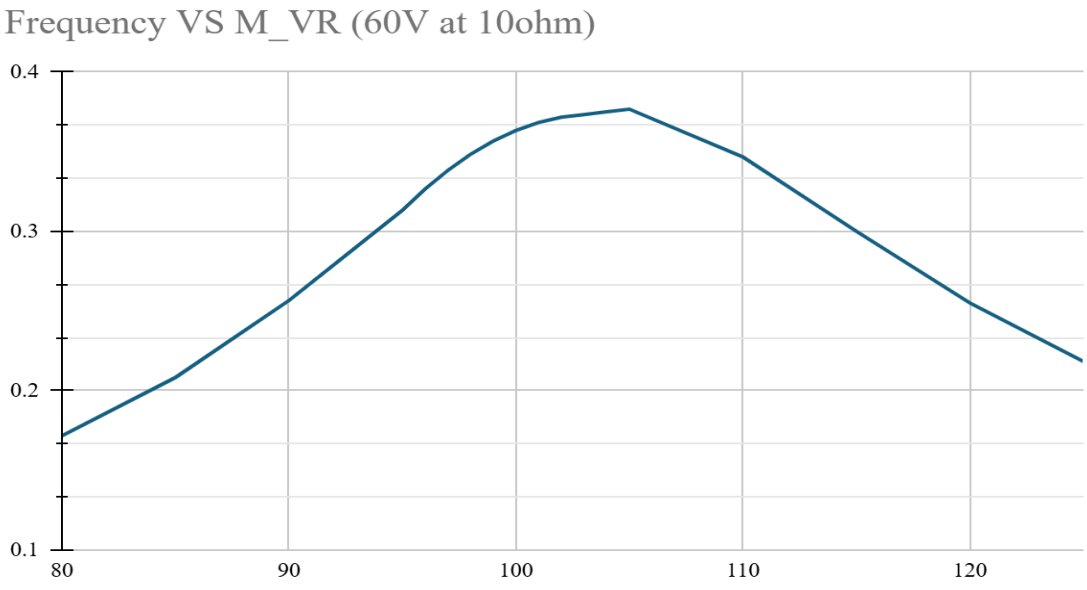


(60V,10Ω)

Input current	Voltage	Current	Frequency	Win	Wout	efficiency	M_VR
0.249	10.3	1.21	80	14.94	12.463	0.8342034806	0.1716666667
0.37	12.5	1.47	85	22.2	18.375	0.8277027027	0.2083333333
0.557	15.4	1.8	90	33.42	27.72	0.829443447	0.2566666667
0.83	18.8	2.21	95	49.8	41.548	0.8342971888	0.3133333333
0.89	19.6	2.3	96	53.4	45.08	0.8441947566	0.3266666667
0.95	20.3	2.38	97	57	48.314	0.8476140351	0.3383333333
1.011	20.9	2.45	98	60.66	51.205	0.8441312232	0.3483333333
1.062	21.4	2.51	99	63.72	53.714	0.8429692404	0.3566666667
1.115	21.8	2.56	100	66.9	55.808	0.834200299	0.3633333333

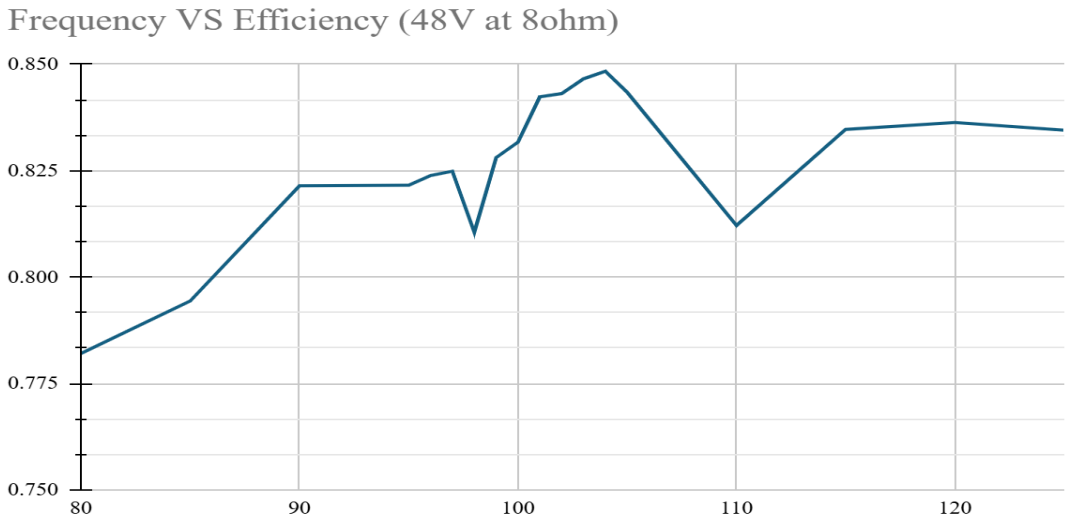
1.153	22.1	2.59	101	69.18	57.239	0.8273923099	0.3683333333
1.182	22.3	2.61	102	70.92	58.203	0.8206852792	0.3716666667
1.193	22.4	2.63	103	71.58	58.912	0.8230231908	0.3733333333
1.196	22.5	2.64	104	71.76	59.4	0.8277591973	0.375
1.189	22.6	2.64	105	71.34	59.664	0.836333053	0.3766666667
1.008	20.8	2.44	110	60.48	50.752	0.8391534392	0.3466666667
0.753	18	2.1	115	45.18	37.8	0.8366533865	0.3
0.548	15.3	1.79	120	32.88	27.387	0.8329379562	0.255
0.405	13.1	1.54	125	24.3	20.174	0.8302057613	0.2183333333

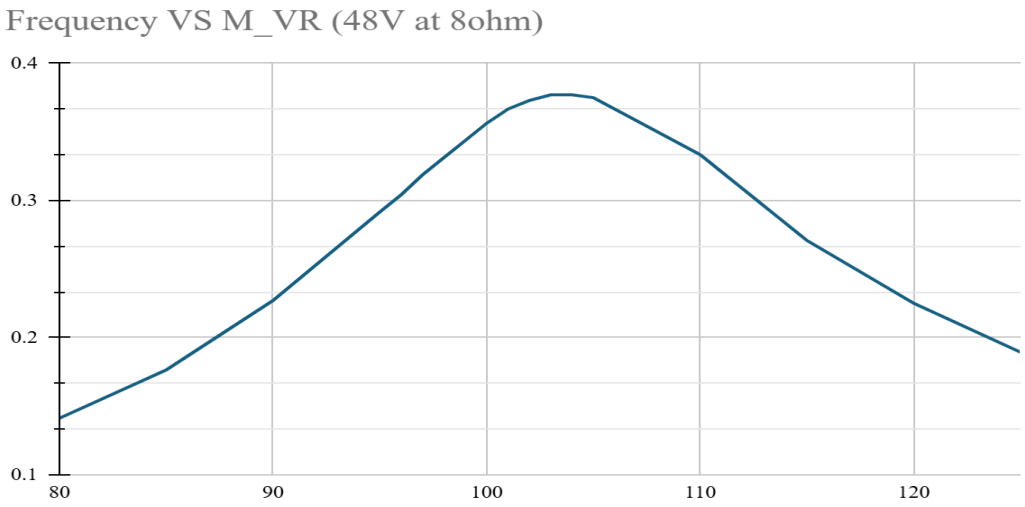




(48V,8Ω)

Input current	Voltage	Current	Frequency	Win	Wout	efficiency	M_VR
0.177	6.78	0.98	80	8.496	6.6444	0.7820621469	0.14125
0.271	8.47	1.22	85	13.008	10.3334	0.7943880689	0.1764583333
0.434	10.9	1.57	90	20.832	17.113	0.8214765745	0.2270833333
0.71	14	2	95	34.08	28	0.8215962441	0.2916666667
0.779	14.6	2.11	96	37.392	30.806	0.8238660676	0.3041666667
0.854	15.3	2.21	97	40.992	33.813	0.824868267	0.31875
0.94	15.9	2.3	98	45.12	36.57	0.8105053191	0.33125
0.988	16.5	2.38	99	47.424	39.27	0.8280617409	0.34375
1.058	17.1	2.47	100	50.784	42.237	0.8316989603	0.35625
1.11	17.6	2.55	101	53.28	44.88	0.8423423423	0.3666666667
1.15	17.9	2.6	102	55.2	46.54	0.843115942	0.3729166667
1.167	18.1	2.62	103	56.016	47.422	0.8465795487	0.3770833333
1.169	18.1	2.63	104	56.112	47.603	0.8483568577	0.3770833333
1.156	18	2.6	105	55.488	46.8	0.8434256055	0.375
0.944	16	2.3	110	45.312	36.8	0.8121468927	0.3333333333
0.61	13	1.88	115	29.28	24.44	0.8346994536	0.2708333333
0.417	10.8	1.55	120	20.016	16.74	0.8363309353	0.225
0.295	9.09	1.3	125	14.16	11.817	0.8345338983	0.189375




Frequency VS M_VR (48V at 8ohm)
0.4
0.3
0.2
0.1
80
90
100
110
120

Task 4: Testing ZVS vs. Hard switching

Aim

In this task, the switching waveforms of a half-bridge operated under ZVS and non-ZVS conditions are observed.

Procedure Conducted

- Start this task by using the setup from Task 3. Set up the oscilloscope to measure the time waveforms of the half-bridge output voltage vinv(t) and the inverter current $i_{inv}(t)$.
- Set the operating frequency to 92.5 kHz. Record and save on a memory stick the screenshots of the vinv(t) and $i_{inv}(t)$ time waveform, including 4-5 periods of the signals. Now zoom in on the signal to focus on the switching interval and save the screenshot again. Turn on the 'ink saver' mode while saving the waveform. The ink-saver mode changes the background color to white.
- Set the frequency of operation to 107.5 kHz and repeat task step ii.

Observations and Results

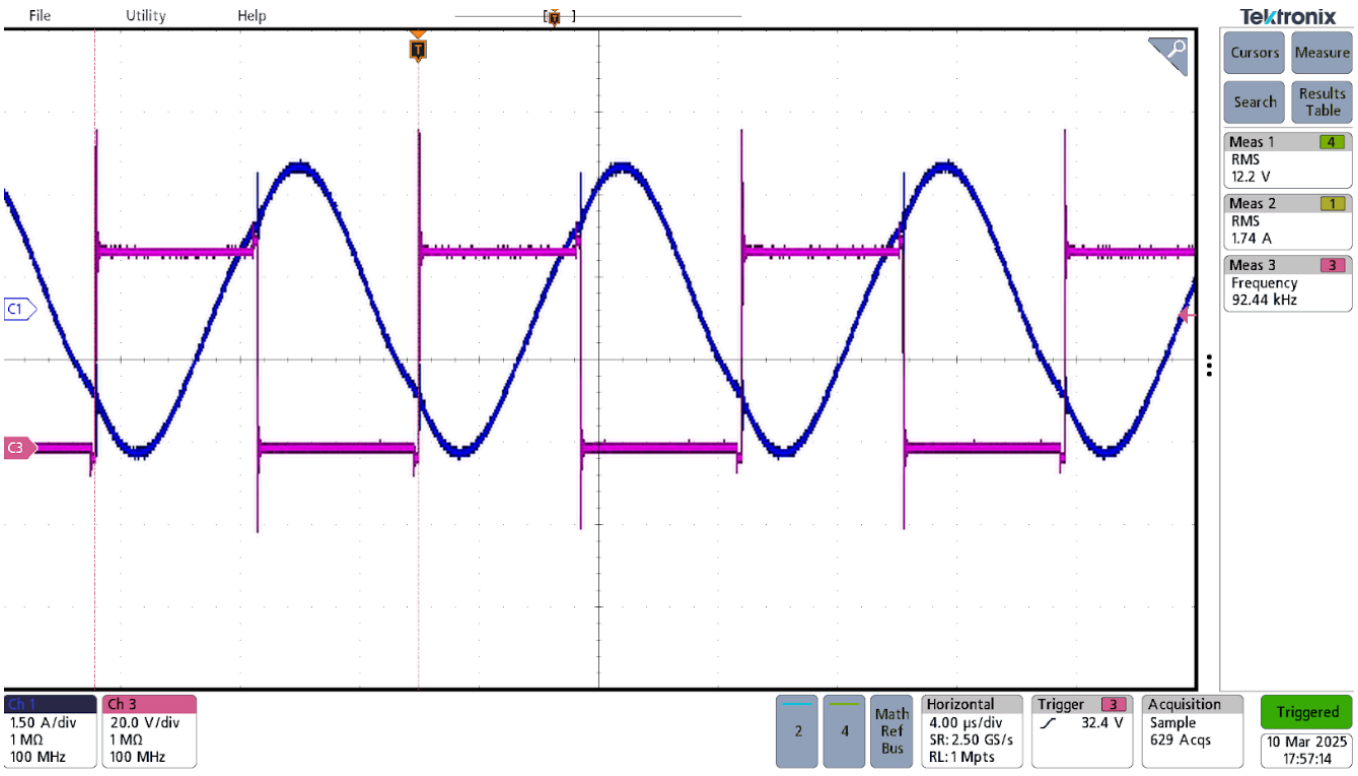


Figure 4.1: Inverter Voltage and Inverter current waveforms for 92.5 KHz

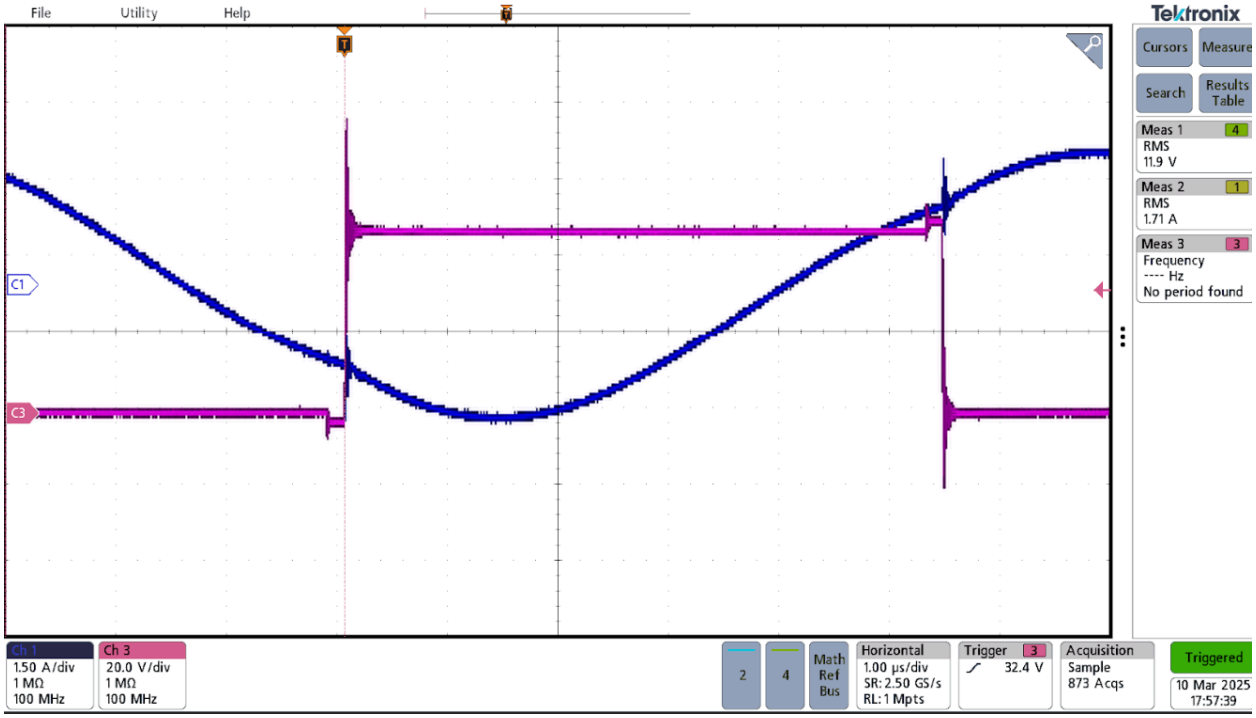


Figure 4.2: Hard switching is observed in this figure. (92.5 KHz)

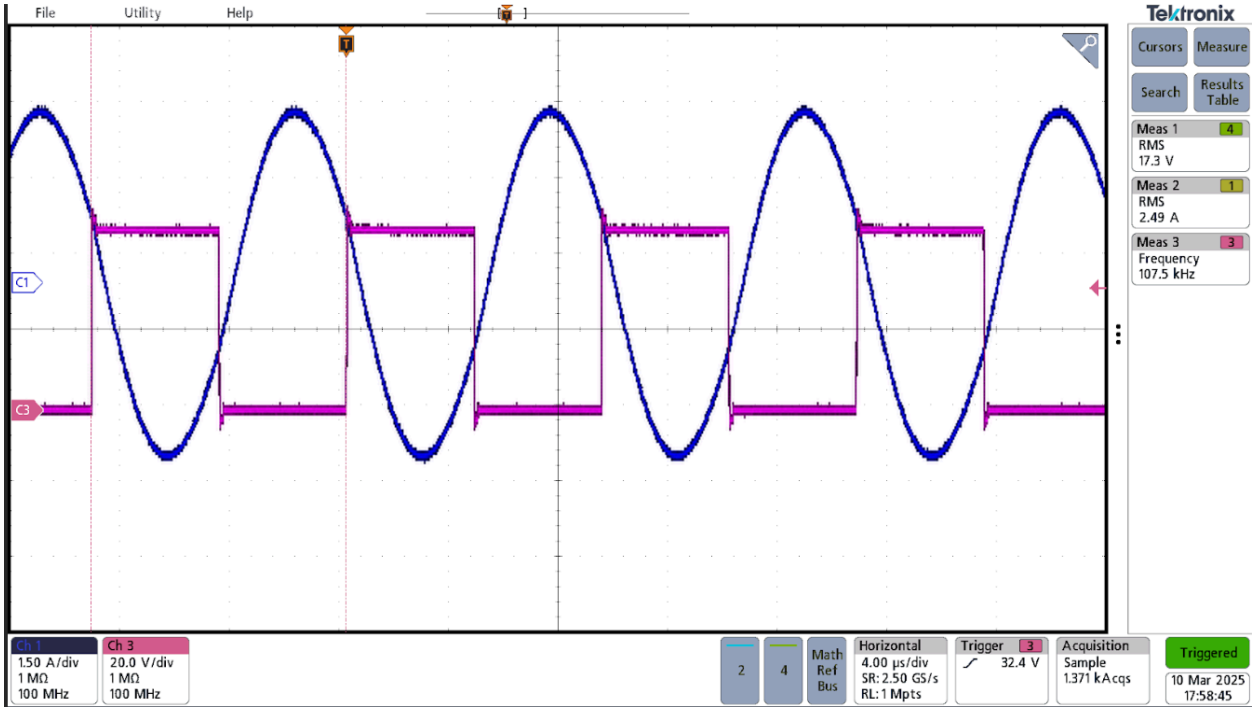


Figure 4.3: Inverter Voltage and Inverter current waveforms for 107.5 KHz

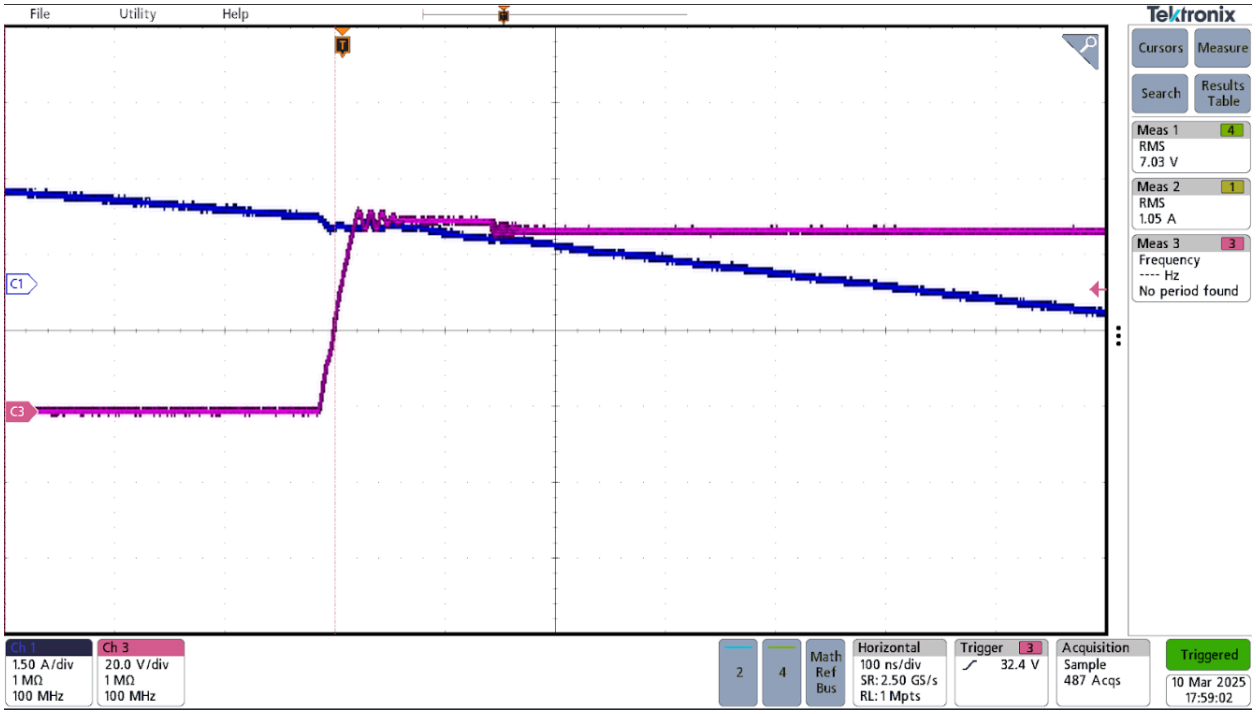


Figure 4.4: ZVS condition is observed (107.5 KHz)

How does the ZVS impact the switching waveforms? What is the general impact of ZVS operation on the entire converter and environment?

ZVS reduces switching losses, which can significantly increase the efficiency of converters, especially at high frequencies and voltages. This is because the energy stored in capacitors is utilized to discharge the switch's output capacitance, allowing for lossless switching transitions EMI is also reduced and component stress is reduced.

Conclusion

In lab 2, the four major tasks were performed in which the primary focus was on the frequency response of passive components. Further, calculating the input resistance values and input inductance values with respect to the changing voltage and the changing resistive loads. The ZVS and hard switching conditions were also observed and recorded.